

UNIVERSITÀ DEGLI STUDI DI TORINO

Corso di Laurea Magistrale in Matematica



UNIVERSITÀ
DI TORINO

Tesi di Laurea Magistrale

The ideally exact categories

A non-pointed generalization
of the notion of semiabelian category

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Anno Accademico 2024-2025

Abstract

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Preface

In this thesis I explore the notion of ideally exact category, introduced by George Janelidze in 2024. The main goal is to provide an overview of the ideally exact categories with the necessary background needed to understand it, and present some of the most important results in this field.

A basic knowledge of category theory — such as limits, colimits, adjunctions, and monads — is assumed throughout this thesis. Some familiarity with Malt'sev, exact, regular, protomodular and semiabelian categories would provide a natural foundation; however the main constructions and definitions of these categories are reviewed in the first chapter. The reader who find it helpful, can also refer to the appendix, where some results about these categories — necessary for proving several results in the thesis — are presented.

Right after the introduction and preliminaries, the second chapter presents the pullback functor and its left adjoint, which are a key tool in the theory of ideally exact categories. In this chapter I also define the connected limits, since they have useful properties with the mentioned adjunction. Then, I will introduce the notion of nullary and essentially nullary monads and their results, which will be used to prove the main result of the ideally exact categories.

The third chapter is devoted to some results concerning the lifting of properties to the Eilenberg-Moore category of a monad. This part is quite technical and articulate. It will be useful to characterize the notion of ideal exact category. After these results, I will present the definition of ideal exact category and some of its properties. Here I will also prove the main result of the ideally exact category, which has been central in motivating interest in this topic.

The fourth chapter is dedicated to the study a monadicity theorem for the pullback functor.

The fifth deals with an overview of the notion of ideal exact category, which is the main topic of this thesis. I will present the definition and the characterization of this notion, as well as the main theorem.

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Introduction

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Chapter 1

Preliminaries

In order to understand the notions introduced in this thesis is important to recall some definitions and constructions. We also take this opportunity to introduce and fix some notations.

First we define some categories that will be used throughout the thesis. Let $\mathcal{A}$ be a category, we defines the following categories.

- The **slice category**: given an object B in a category $\mathcal{A}$, we call $(\mathcal{A} \downarrow B)$ the category of objects over B , we denote an object by $(A, \alpha : A \rightarrow B)$. A morphism between two objects (A, α) and (A', α') is a morphism $f : A \rightarrow A'$ such that $\alpha' f = \alpha$.

$$\begin{array}{ccc} A & & A \xrightarrow{f} A' \\ \downarrow \alpha & & \downarrow \alpha \swarrow \alpha' \\ B & & B \end{array}$$

- The **coslice category**: given an object B in a category $\mathcal{A}$, we call $(B \downarrow \mathcal{A})$ the category of objects under B , we denote an object by $(\alpha : B \rightarrow A, A)$. A morphism between two objects (α, A) and (α', A') is a morphism $f : A \rightarrow A'$ such that $f\alpha = \alpha'$.

$$\begin{array}{ccc} B & & B \\ \downarrow \alpha & & \downarrow \alpha \searrow \alpha' \\ A & & A \xrightarrow{f} A' \end{array}$$

- The **category of points**: given an object B in a category $\mathcal{A}$, we call $\text{Pt}(B)$ the category of points of B , we denote an object by (A, B, p, s) which is

a diagram of the type $A \begin{smallmatrix} \xrightarrow{p} \\ \xleftarrow{s} \end{smallmatrix} B$ with the condition $ps = 1_B$. Morphisms between two objects (A, B, p, s) and (A', B', p', s') are diagrams of the type

$$\begin{array}{ccc} A & \xrightarrow{f} & A' \\ \swarrow s & & \searrow p' \\ & B & \\ \searrow p & & \swarrow s' \end{array}$$

with the conditions $p'f = p$ and $fs = s'$

We recall the following definitions.

Definition 1.0.1. A finitely complete category $\mathcal{A}$ is said to be **regular** if

- Every morphism in $\mathcal{A}$ admits an (regular epi, mono) factorization;
- Regular epis in $\mathcal{A}$ are pullback stable.

Definition 1.0.2. A category $\mathcal{A}$ is said to be **exact** if

- It is regular;
- Any equivalence relation is the kernel pair of some arrow.

In order to define the next notion, we need the following construction. Suppose $\mathcal{A}$ has pullbacks and consider a morphism $f : B' \rightarrow B$ in the same category. We define the **change-of-base functor** of f .

$$f^* : \text{Pt}(B) \rightarrow \text{Pt}(B')$$

$$\begin{array}{ccc} \begin{array}{c} A \\ \uparrow s \quad \downarrow p \\ B \end{array} & \xrightarrow{f^*} & \begin{array}{ccccc} & & B' & & \\ & \searrow \langle 1_{B'}, sf \rangle & \uparrow sf & & \\ & & B' \times_B A & \xrightarrow{\text{proj}_2} & A \\ & \uparrow 1_{B'} & \downarrow \text{proj}_1 & & \uparrow s \quad \downarrow p \\ & & B' & \xrightarrow{f} & B \end{array} \end{array}$$

where $\langle 1_{B'}, sf \rangle$ is the section of proj_1 . It holds that $(\text{proj}_1)\langle 1_{B'}, sf \rangle = 1_{B'}$. The action on a morphism $\alpha : A \rightarrow A'$ is given by

The diagram illustrates the relationship between a morphism $\alpha: A \rightarrow A'$ and its pullback $f^*: \text{Pt}(B) \rightarrow \text{Pt}(A)$. On the left, a square shows A and A' at the top, B at the bottom, with a vertical arrow $s: A \rightarrow B$ and a diagonal arrow $p: A \rightarrow B$. A morphism $\alpha: A \rightarrow A'$ is shown at the top, and a diagonal arrow $s': A' \rightarrow B$ is shown at the bottom. A horizontal arrow f^* points from this square to a larger, more complex diagram on the right. The right diagram shows a pullback square with $B' \times_B A$ and $B' \times_B A'$ at the top, B at the bottom, and B' at the bottom. The top horizontal arrow is α . The left vertical arrow is proj_2 , and the right vertical arrow is proj'_2 . The bottom horizontal arrow is f . The left diagonal arrow is p , and the right diagonal arrow is p' . The bottom-left diagonal arrow is proj_1 , and the bottom-right diagonal arrow is proj'_1 . The bottom-left curved arrow is $\langle 1_{B'}, sf \rangle$, and the bottom-right curved arrow is $\langle 1_{B'}, s'f \rangle$. A dashed curved arrow at the top is labeled $f^*(\alpha)$.

Definition 1.0.3. A finitely complete category $\mathcal{A}$ is said to be **protomodular** if for every morphism $f: A \rightarrow B$ in $\mathcal{A}$, the change-of-base functor $f^*: \text{Pt}(B) \rightarrow \text{Pt}(A)$ is conservative.

We conclude recalling the following.

Definition 1.0.4. A category finitely complete category $\mathcal{A}$ is **Mal'tsev** if every internal reflexive relation in $\mathcal{A}$ is an equivalence relation.

Definition 1.0.5. A category $\mathcal{A}$ is said to be **semiabelian** if

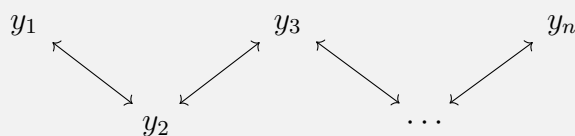
- It is pointed;
- It is exact;
- It is protomodular;
- It has binary coproducts.

Chapter 2

Pullback functor

2.1 Connected limits

Definition 2.1.1. A **connected limit** is a limit over a connected category. A category is connected if it is inhabited and any two objects can be connected by a zigzag of morphisms



where each symbol $\leftrightarrow$ is a morphism $y_i \rightarrow y_{i+1}$ or $y_{i+1} \rightarrow y_i$.

Example 2.1.2. Some examples

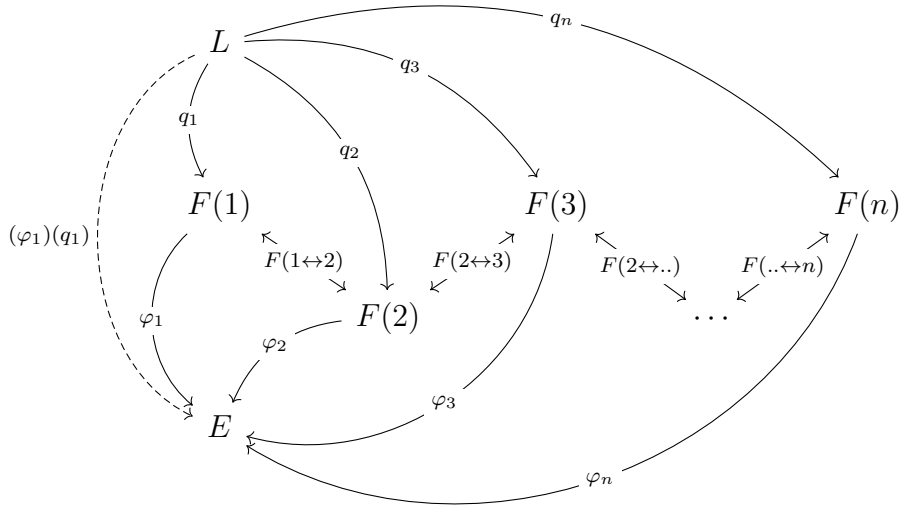
- The pullback is a connected limit.

$$\begin{array}{ccc}
 & & 2 \\
 & & \downarrow j \\
 1 & \xrightarrow{i} & 3
 \end{array}$$

- The equalizer is a connected limit.

$$1 \begin{array}{c} \xrightarrow{i} \\ \xrightarrow{j} \end{array} 2$$

Remark 2.1.3. *The connected limits in the category $(\mathcal{A} \downarrow E)$ can be computed in $\mathcal{A}$. In fact we can provide to the object L of the limit cone $(L, (q_i)_{i \in \mathcal{I}})$ in $\mathcal{A}$ a morphism to form an object in $(\mathcal{A} \downarrow E)$. In the zigzag below we chose $(\varphi_1)(q_1)$, but we could use whatever morphism of the type $(\varphi_i)(q_i)$. The connection guarantees that all the morphisms $q_1 \dots q_n$ involved in the limit in $\mathcal{A}$ are also morphisms in $(\mathcal{A} \downarrow E)$, ie $(\varphi_i)(q_i) = (\varphi_1)(q_1) \forall i$.*



By the previous result, the following holds.

Proposition 2.1.4. *Let E be an object in $\mathcal{A}$. Then the forgetful functor $U : (\mathcal{A} \downarrow E) \rightarrow \mathcal{A}$ creates all connected limits that $\mathcal{A}$ has.*

Note that this result is not true for all the limits. This is extensively shown in the following example.

Example 2.1.5. *The product is not a connected limit, in fact as we can observe the following category is not connected.*

1 2

We try to compute the product directly in $(\mathcal{A} \downarrow E)$. Let (D, δ) and (D', δ')

be two objects in $(\mathcal{A} \downarrow E)$, the product is given by the following pullback

$$\begin{array}{ccccc}
 & D \times_B D' & \longrightarrow & D' & \\
 & \swarrow & & \searrow \delta' & \\
 D & \xrightarrow{\delta} & E & & \\
 & \searrow \delta & \downarrow 1_E & \swarrow \delta' & \\
 & & E & &
 \end{array}$$

which doesn't correspond to the product $D \times D'$ in $\mathcal{A}$.

2.2 Pullback functor's adjunction

Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow B$ in $\mathcal{A}$, one can consider functors pullback and shrink

$$\begin{array}{ccc}
 & p^* & \\
 (\mathcal{A} \downarrow B) & \xrightarrow{\quad} & (\mathcal{A} \downarrow E) \\
 & p_! &
 \end{array}$$

defined as follows:

- $p^* : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the **pullback functor**

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & & \\ \downarrow \text{proj}_1 & & \downarrow \alpha & & \\ E & \xrightarrow{p} & B & & \end{array} \\
 \\
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \swarrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} & & p^*(f) = \tilde{f} & & \\ & \xrightarrow{\quad} & \xrightarrow{\quad} & \xrightarrow{\quad} & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & & \swarrow \alpha' & & & \\ & & B & & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}'_1 & & \\ & E & & & & & \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the **shrink functor**

$$\begin{array}{ccc}
\begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{c} D \\ \delta \downarrow \quad \searrow p\delta \\ E \xrightarrow{p} B \end{array} \\
\begin{array}{ccc} D & \xrightarrow{g} & D' \\ \delta \downarrow & \nearrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
\end{array}$$

Proposition 2.2.1. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then $p_!$ preserves connected limits.*

Proof. By Remark 2.1.3 connected limits are computed in $\mathcal{A}$ and $p_!$ doesn't affect objects and morphism in $\mathcal{A}$. $\square$

Proposition 2.2.2. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks, then*

$$\begin{array}{ccc}
& p^* & \\
(\mathcal{A} \downarrow B) & \xrightarrow{\quad} & (\mathcal{A} \downarrow E) \\
& p_! &
\end{array} \quad \text{with } p_! \dashv p^*$$

is an adjunction. The unit is given by

$$\begin{array}{l}
\eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p_! \\
\eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)
\end{array}$$

$$\begin{array}{ccccc}
D & & & & D \\
& \searrow \langle \delta, 1_D \rangle & & \searrow 1_D & \\
& E \times_B D & \xrightarrow{\quad} & D & \\
\delta \downarrow & \downarrow & & \downarrow p\delta & \\
E & \xrightarrow{p} & B & &
\end{array}$$

and the counit is given by

$$\begin{array}{l}
\epsilon : p_! p^* \rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
\epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)
\end{array}$$

$$\begin{array}{ccc}
E \times_B A & \xrightarrow{\text{proj}_2} & A \\
\downarrow \text{proj}_1 & & \downarrow \alpha \\
E & \xrightarrow{p} & B
\end{array}$$

Proof. We want to prove this is adjunction. We start proving that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc} \begin{array}{ccc} D & \xrightarrow{g} & D' \\ \downarrow \delta & \searrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ & & B \end{array} \\ \\ \xrightarrow{p^*} & & \begin{array}{c} \xrightarrow{\tilde{g}} \\ \begin{array}{ccccc} E \times_B D & \xrightarrow{\text{proj}_2} & D & \xrightarrow{g} & D' & \xleftarrow{\text{proj}'_2} & E \times_B D' \\ \nwarrow \eta_{(D, \delta)} & \searrow p\delta & \searrow p\delta' & \nwarrow \eta_{(D', \delta')} & \\ & B & & & \\ \nwarrow \text{proj}_1 & \uparrow p & \nearrow \text{proj}'_1 & & \\ & E & & & \end{array} \end{array} \end{array}$$

Consider the following commutative diagram.

$$\begin{array}{ccccc} D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D & & \\ \downarrow g & \searrow & \downarrow p^* p_!(g) = \tilde{g} & & \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\ & & E & \xrightarrow{p} & B \end{array}$$

Since $(\text{proj}'_1, \text{proj}'_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D, \delta)}) = (g)(\text{proj}_2)(\eta_{(D, \delta)}) = (g)(1_D) = g$

- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(g) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. Thus the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\ p!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A' \end{array}$$

As similarly done above, we need to expand the action of the functor $p!p^*$ on the morphism f , we have:

$$\begin{array}{c} \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \nearrow \alpha' & \\ B & & \end{array} \xrightarrow{p^*} \begin{array}{ccccc} & & \tilde{f} & & \\ & & \text{---} & & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & \searrow \alpha' & & \\ & & B & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \swarrow \text{proj}'_1 \\ & & E & & \end{array} \\ \\ \xrightarrow{p!} \begin{array}{ccccc} & & \tilde{f} & & \\ & & \text{---} & & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & \searrow \alpha' & & \\ & & B & & \\ p(\text{proj}_1) \swarrow & & & & \swarrow p(\text{proj}'_1) \end{array} \end{array}$$

Since $\epsilon_{(A,\alpha)} = \text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. Thus also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$.

$$((p^* \epsilon) \circ (\eta p^*))_{(A, \alpha)} = (p^* \epsilon)_{(A, \alpha)} (\eta p^*)_{(A, \alpha)} = p^* (\text{proj}_{2, E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1, E \times_B A}$

$$\begin{array}{c}
 \begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A \\
 \searrow \alpha(\text{proj}_{2, E \times_B A}) & & \swarrow \alpha \\
 & B &
 \end{array}
 \xrightarrow{p^*}
 \end{array}$$

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & \text{proj}_{2, E \times_B A} & & \\
 & \text{proj}_{2, E \times_B A} & & \text{proj}_{2, E \times_B A} & \\
 E \times_B (E \times_B A) & \xrightarrow{\text{proj}_{2, E \times_B (E \times_B A)}} & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A & \xleftarrow{\text{proj}_{2, E \times_B A}} & E \times_B A \\
 & & \searrow (\alpha)(\text{proj}_{2, E \times_B A}) & & \swarrow \alpha \\
 & & & B & \\
 & \text{proj}_{1, E \times_B (E \times_B A)} & & \uparrow p & \text{proj}_{1, E \times_B A} \\
 & & E & &
 \end{array}
 \end{array}$$

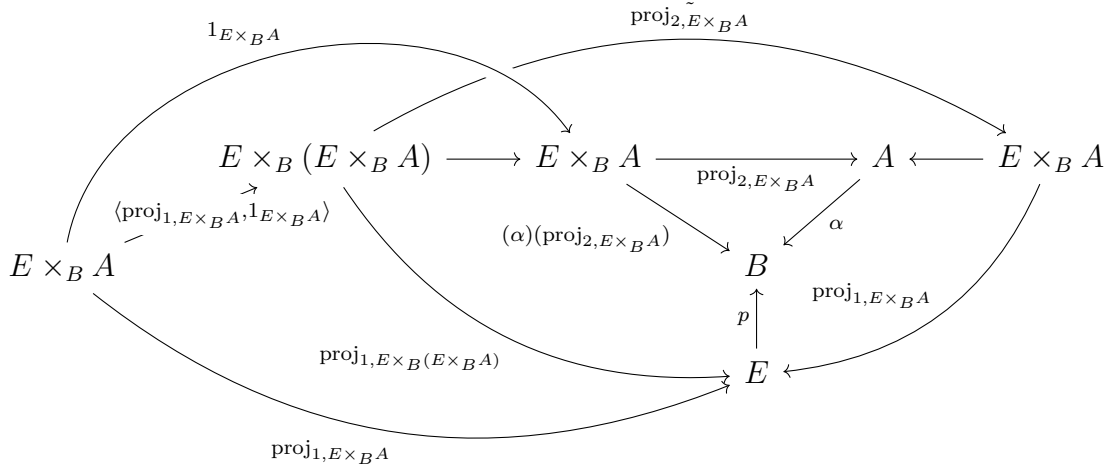
$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A}) (\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of η , we obtain

$$\begin{array}{c}
 \begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \text{proj}_{1, E \times_B A} & & \text{proj}_{1, E \times_B A} & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \xrightarrow{\quad} & E \times_B A \\
 \searrow \text{proj}_{1, E \times_B A} & & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) \\
 & E & \xrightarrow{p} & B &
 \end{array}
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A}) \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.



We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $1_{(E \times_B A, \text{proj}_1)}$. Then, since the projections are jointly monic, we have the result. Following the first composition

$$\begin{aligned}
 & (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = \\
 & = (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) \\
 & = (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = (\text{proj}_{2, E \times_B A})(1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$\begin{aligned}
 & (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = \\
 & = (\text{proj}_{1, E \times_B (E \times_B A)})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) \\
 & = \text{proj}_{1, E \times_B A} \\
 & = (\text{proj}_{1, E \times_B A})(1_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Hence

$$\circledast = (\tilde{\text{proj}}_{2, E \times_B A})(\text{proj}_{1, E \times_B A}, 1_{E \times_B A}) = 1_{(E \times_B A, \text{proj}_1)}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p_!) \circ (p_! \eta))_{(D, \delta)} &= ((\epsilon p_!)_{(D, \delta)} (p_! \eta))_{(D, \delta)} = (\epsilon_{p_!(D, \delta)}) p_! \langle \delta, 1_D \rangle = \\ &= (\epsilon_{(D, p\delta)}) p_! (\langle \delta, 1_D \rangle) = \text{proj}_{2, E \times_B D} \langle \delta, 1_D \rangle = 1_{(D, p\delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds. $\square$

Remark 2.2.3. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

Property 2.2.4. Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$. The diagram constructed by p^* acting on a morphism $f : (A, \alpha) \rightarrow (A', \alpha')$ is a pullback in $(\mathcal{A} \downarrow B)$.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 \\ A & \xrightarrow{f} & A' \end{array}$$

Proof. We prove it in $\mathcal{A}$, then by Remark 2.1.3 it holds in $\mathcal{A} \downarrow B$. Since the right and the total diagrams below are pullback, then the left diagram is a pullback.

$$\begin{array}{ccccc} & & \text{proj}_1 & & \\ & \nearrow & & \searrow & \\ E \times_B A & \xrightarrow{p^*(f)=\tilde{f}} & E \times_B A' & \xrightarrow{\text{proj}'_1} & E \\ \downarrow \text{proj}_2 & & \downarrow \text{proj}'_2 & & \downarrow p \\ A & \xrightarrow{f} & A' & \xrightarrow{\alpha'} & B \\ & \searrow & & \nearrow & \\ & & \alpha & & \end{array}$$

$\square$

Property 2.2.5. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then the unit η and the counit ϵ of the adjunction associated to p are cartesian.*

Proof. First we prove that η is cartesian. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Applying *Property 2.2.4* to $p_!(g)$, we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow p^*p_!(g)=\tilde{g} & & \downarrow g \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \end{array}$$

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc} & & 1_D & & \\ & \nearrow & & \searrow & \\ D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow g & & \downarrow p^*p_!(g)=\tilde{g} & & \downarrow g \\ D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & \nwarrow & & \nearrow & \\ & & 1_{D'} & & \end{array}$$

Now we want to prove that ϵ is cartesian. Let $f : (A, \alpha) \rightarrow (A', \alpha')$ be a morphism in $(\mathcal{A} \downarrow B)$. By *Proposition 2.2.1* applied at the pullback in *Property 2.2.4* for f we obtain that the following diagram is a pullback.

$$\begin{array}{ccc} E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}=\text{proj}_2} & A \\ \downarrow p_!p^*(f)=\tilde{f} & & \downarrow f \\ E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}=\text{proj}'_2} & A' \end{array}$$

□

Proposition 2.2.6. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then p^* preserves coequalizer of kernel pairs whose coequalizers are pullback*

stable regular epimorphisms.

Proof. Consider the following diagram:

$$\begin{array}{ccccc}
 p^*(\text{Eq}(q)) & \xrightleftharpoons[p_1]{\tilde{p}_2} & p^*(A) = A \times_B E & \xrightarrow{\tilde{q}} & p^*(\text{Coeq}(p_1, p_2)) \\
 \downarrow & & \downarrow & & \downarrow \\
 \text{Eq}(q) & \xrightleftharpoons[p_1]{p_2} & A & \xrightarrow{q} & \text{Coeq}(p_1, p_2)
 \end{array}$$

Let q be the coequalizer of the kernel pair $\text{Eq}(q)$. We can affirm by *Property 2.2.4* applied to q that the diagram in the bottom-right corner is a pullback. Since, by hypothesis, q is a pullback stable regular epimorphism, we have that $\tilde{q}$ is a regular epimorphism.

We now aim to show that $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. Since $\tilde{q}$, $\tilde{p}_1$ and $\tilde{p}_2$ are obtained applying p^* to the kernel pair $\text{Eq}(q)$, and p^* preserves limits, it follows that $p^*(\text{Eq}(q))$ is the kernel pair of $\tilde{q}$. We already know that $\tilde{q}$ is a regular epi. Thus, by *Proposition A.0.6*, $\tilde{q}$ is also the coequalizer of $(\tilde{p}_1, \tilde{p}_2)$. $\square$

Later on it will be useful to study monads. Thus, we state the following proposition.

Property 2.2.7. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ with pullbacks. Then the associated monad $(p^*p_!, \eta, p^*\epsilon p_!)$ is cartesian.*

Proof. We need to prove that the category $(\mathcal{A} \downarrow E)$ has pullback, the monad preserves pullbacks, and both the multiplication $p^*\epsilon p_!$ and the unit η are cartesian.

The category $(\mathcal{A} \downarrow E)$ has pullbacks, by *Proposition 2.1.4*, since $\mathcal{A}$ has. The monad $p^*p_!$ preserves pullbacks, in fact p^* is a right adjoint so it preserves limits, therefore preserves pullbacks, and $p_!$ preserves pullbacks by *Proposition 2.2.1*. We already proved that the unit η is cartesian, so it remains to prove that the multiplication $p^*\epsilon p_!$ is cartesian. Since ϵ is cartesian and p^* preserves pullbacks as shown above, we can conclude that the multiplication is cartesian. $\square$

2.3 Nullary and essentially nullary monads

In this section we look for conditions such that the monad associated to a morphism $p : E \rightarrow B$ is essentially nullary. This will be useful in the main theorem of the ideally exact categories. This section is based on the work of [Jan24].

Definition 2.3.1. A monad (T, η, μ) on a category $\mathcal{X}$ with finite coproducts is said to be **nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is an iso, where $!_X$ is the unique morphism from the initial object of $\mathcal{X}$ to X . A monad on a category $\mathcal{X}$ with finite coproducts is said to be **essentially nullary** if, for each object X in $\mathcal{X}$, the morphism $[T(!_X), \eta_X]$ is a strong epi.

The following lemma, from [BB04], will be useful for our purposes.

Lemma 2.3.2. Let $\mathcal{X}$ be a protomodular category. Then for each commutative diagram of the form

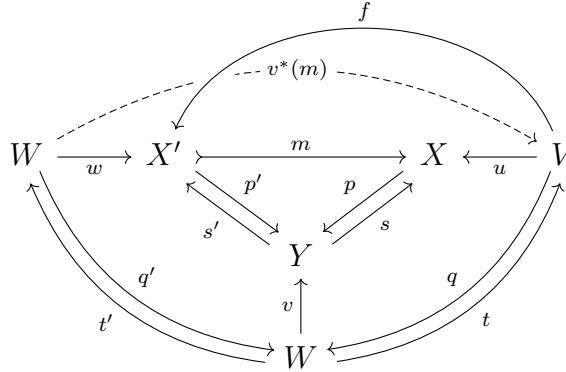
$$\begin{array}{ccc} V & \xrightarrow{u} & X \\ q \downarrow \uparrow t & & p \downarrow \uparrow s \\ W & \xrightarrow{v} & Y \end{array}$$

where p, q are split epimorphisms with respective sections s, t and the downward directed square is a pullback, it holds that the pair (u, s) is a strongly epimorphic pair.

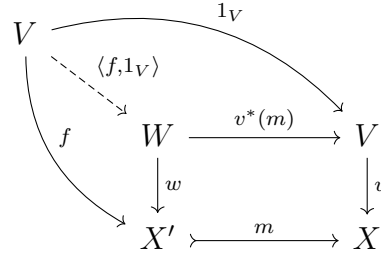
Proof. Assume $\mathcal{X}$ is protomodular. We want to prove that if u and s factorize through a mono m as $u = mf$ and $s = ms'$, then m is an iso, which means (u, s) is an extremally epimorphic pair. By a standard result, proved in *Proposition A.0.4*, this implies that (u, s) is a strongly epimorphic pair.

Define $p' = pm$, we have that $p's' = 1_Y$, in fact $p's' = pms' = ps = 1_Y$. Pulling back with the change-of-base functor v^* the mono m we obtain the following diagram. In this operation we also generate the morphisms q', t' and w . In the same way as in *Property 2.2.4* we have that the diagram

between m and $v^*(m)$ is a pullback.



Now we want to prove that $v^*(m)$ is an iso. Consider the following diagram. We know that the external diagram commutes by the hypothesis factorization.



Since the square is a pullback, it's obvious that $v^*(m)\langle f, 1_V \rangle = 1_V$. It remains to prove that $\langle f, 1_V \rangle v^*(m) = 1_W$.

Notice that $mf v^*(m) = uv^*(m) = mw$, so $f v^*(m) = w$ since m is a mono. Now consider the following compositions, and use what we just noticed.

- $v^*(m)(\langle f, 1_V \rangle v^*(m)) = (v^*(m)\langle f, 1_V \rangle)v^*(m) = 1_V v^*(m) = v^*(m)1_W$
- $w(\langle f, 1_V \rangle v^*(m)) = (w\langle f, 1_V \rangle)v^*(m) = f v^*(m) = w = w1_W$

Since the pair $(v^*(m), w)$ is jointly monic we have that $\langle f, 1_V \rangle v^*(m) = 1_W$. Considering this composition and the one above, it follows that $v^*(m)$ is an iso.

By the protomodularity of $\mathcal{X}$, since $v^*(m)$ is an iso, we have that m is an iso, as claimed. $\square$

We now aim to prove a sufficient condition for the monad associated to a morphism $p : E \rightarrow B$ to be essentially nullary. We could start with the

following proposition.

Theorem 2.3.3. *If $\mathcal{X}$ is a pointed protomodular category with finite coproducts and (T, η, μ) is a monad on $\mathcal{X}$ with cartesian η , then the monad is essentially nullary.*

Proof. For each object X in $\mathcal{X}$ we can consider the following commutative diagram:

$$\begin{array}{ccc} X & \xrightarrow{\eta_X} & T(X) \\ \downarrow !^X & \uparrow !_X & \downarrow T(!^X) \\ 0 & \xrightarrow{\eta_0} & T(0) \end{array}$$

By hypothesis η is cartesian, so the downward directed square is a pullback. Since $\mathcal{X}$ is protomodular then by *Lemma 2.3.2*, the pair $(\eta_X, T(!^X))$ is strongly epimorphic. By a standard result, proved in *Proposition A.0.5*, this yields that $[\eta_X, T(!^X)]$ is a strong epi. Thus the monad is essentially nullary. $\square$

In our case $A \downarrow E$ is pointed if and only if E is a an initial objects, in fact in this case the following initial and terminal objects coincide.

$$\begin{array}{ccc} E & \xrightarrow{!_A} & A \\ & \searrow & \downarrow \alpha \\ & & E \end{array} \quad \begin{array}{ccc} A & \xrightarrow{\alpha} & E \\ \downarrow \alpha & \swarrow 1_E & \\ E & & \end{array}$$

While this would work in that specific case, we would like to drop such condition. The following result fits our needs.

Theorem 2.3.4. *Let $p : E \rightarrow B$ be a morphism in $\mathcal{A}$ protomodular with finite coproducts. Then the monad associated to p is essentially nullary.*

Proof. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$. To prove that the monad is essentially nullary we need to study the morphism of the initial object. So first note the morphism for each object (D, δ) in $(\mathcal{A} \downarrow E)$ the morphism of the initial object $!_D$ is the same as the morphism of the initial object in

$\mathcal{A}$, in fact the following diagram commutes.

$$\begin{array}{ccc} 0 & \xrightarrow{!_D} & D \\ & \searrow \scriptstyle !_E & \downarrow \scriptstyle \delta \\ & & E \end{array}$$

Then consider the following diagram.

$$\begin{array}{ccc} E \times_B 0 & \xrightarrow{p^*p_!(!_D)} & E \times_B D \\ \text{proj}_2 \downarrow \uparrow \langle !_E, 1_0 \rangle & & \text{proj}'_2 \downarrow \uparrow \langle \delta, 1_D \rangle \\ 0 & \xrightarrow{!_D} & D \end{array}$$

The downward directed square is commutative by the definition of p^* , and it is a pullback by *Property 2.2.4*. The upward directed square commutes by naturality of η , in fact $\eta_{(0,!_E)} = \langle !_E, 1_0 \rangle$ and $\eta_{(D,\delta)} = \langle \delta, 1_D \rangle$. So, by *Lemma 2.3.2*, the pair $(p^*(!_D), \langle !_E, 1_0 \rangle)$ is strongly epimorphic. By *Proposition A.0.5* this means that $[T(!_D), \eta_{(D,\delta)}] = [p^*p_!(!_D), \langle \delta, 1_D \rangle]$ is a strong epi. Thus the monad is essentially nullary. $\square$

Chapter 3

Lifting results on the E-M categories

3.1 Exactness and protomodularity lift

We present here some technical results on the Eilenberg-Moore categories. These will play a key role in the definition and characterization of the ideally exact categories.

Proposition 3.1.1. *Let T be a monad preserving regular epimorphisms and kernel pairs. If $\mathcal{X}$ is exact, then the Eilenberg-Moore category $\mathcal{X}^T$ is exact.*

Proof. Assume $\mathcal{X}$ is exact. With the given hypotheses on the monad T , we aim to prove that $\mathcal{X}^T$ is exact. We need to establish that it is finitely complete, that each equivalent relation is the kernel pair of some arrow, that regular epis are pullback stable and that each morphism admits a (regular epi, mono) factorization.

First we show that $\mathcal{X}^T$ is finitely complete. We recall that the forgetful functor $U^T : \mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has. Since $\mathcal{X}$ is exact, it has all finite limits. Therefore, $\mathcal{X}^T$ has all finite limits.

We proceed to prove that each equivalent relation is the kernel pair of some arrow. Consider an equivalence relation R in $\mathcal{X}^T$ with ξ_R and ξ_A giving the structure of algebras, as in the diagram below. Then, under the forgetful functor U^T , R is an equivalence relation in $\mathcal{X}$. $\mathcal{X}$ is exact, so R is the kernel pair of some arrow. In particular, by *Proposition A.0.9*, we can suppose that R is the kernel pair of its coequalizer. Let call it $q : A \rightarrow C$.

Since T preserves kernel pairs, we have that $T(R)$ is the kernel pair of $T(q)$. Moreover T preserves regular epis, so $T(q)$ is a regular epi. By *Proposition A.0.6*, we can assume that $T(q)$ is the coequalizer of its kernel pair $(T(R), T(r_1), T(r_2))$. Since $q\xi_A T(r_1) = qr_1\xi_R = qr_2\xi_R = q\xi_A T(r_2)$, by the universal property of the coequalizer, there exists a unique morphism ξ_C such that $\xi_C T(q) = q\xi_A$. The morphism ξ_C is the algebra structure of $T(B)$. Since U^T creates all the limits that $\mathcal{X}$ has, we can conclude that R is the kernel pair in $\mathcal{X}^T$ of the arrow q .

$$\begin{array}{ccccc}
 T(R) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(q)} & T(C) \\
 \downarrow \xi_R & & \downarrow \xi_A & & \downarrow \xi_C \\
 R & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{q} & C
 \end{array}$$

Consider now a morphism $f : (A, \xi_A) \rightarrow (B, \xi_B)$ in $\mathcal{X}^T$, we aim to prove that it admits a (regular epi, mono) factorization. The morphism f , under the forgetful functor U^T , is a morphism in $\mathcal{X}$, which is exact and in particular regular. Thus f admits a (regular epi, mono) factorization in $\mathcal{X}$. Let it be $f = me$ with $m : Q \rightarrow B$ mono and $e : A \rightarrow Q$ regular epi, thus the coequalizer of some pair of parallel arrows. A and B already have an algebra structure. We want to provide Q and the domain of the parallel arrows with a structure of algebra, so that we can lift the factorization to $\mathcal{X}^T$.

$$\begin{array}{ccccc}
 T(X) & \xrightleftharpoons{\quad} & T(A) & \xrightarrow{T(f)} & T(B) \\
 & & \downarrow \xi_A & \searrow T(e) & \nearrow T(m) \\
 & & & T(Q) & \\
 & & & \downarrow \xi_B & \\
 X & \xrightleftharpoons{\quad} & A & \xrightarrow{f} & B \\
 & & \searrow e & & \nearrow m \\
 & & & Q &
 \end{array}$$

In order to do so, we first need to show that T preserves coequalizers in $\mathcal{X}$ of kernel pairs arising from $\mathcal{X}^T$, once the algebra structure is forgotten, and considering only the pair of the whole kernel pair. Let (Eq, r_1, r_2) be a kernel pair in $\mathcal{X}$, arising from a kernel pair in $\mathcal{X}^T$ of a morphism, and let p be the coequalizer in $\mathcal{X}$ of (r_1, r_2) . By *Proposition A.0.7*, we can affirm

that (r_1, r_2) is the kernel pair of p . Given that (Eq, r_1, r_2) is a kernel pair arising from $\mathcal{X}^T$, it follows that $(T(\text{Eq}), T(r_1), T(r_2))$ is the kernel pair of some arrow. Since T preserves regular epi and kernel pair we can assert that $T(p)$ is a regular epi and, by *Proposition A.0.6*, it is the coequalizer of the kernel pair $(T(\text{Eq}), T(r_1), T(r_2))$, as claimed.

$$\begin{array}{ccccc}
 T(\text{Eq}) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(p)} & T(B) \\
 \downarrow \xi_{\text{Eq}} & & \downarrow \xi_A & & \downarrow \xi_B \\
 \text{Eq} & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{p} & B
 \end{array}$$

Picking up where we left off, consider f and its kernel pair $(\text{Eq}(f), r_1, r_2)$. Since U^T creates all the limits that $\mathcal{X}$ has, we can lift the kernel pair $(\text{Eq}(f), r_1, r_2)$ to a kernel pair in $\mathcal{X}^T$. Let ξ_{Eq} be the structure of algebra of Eq .

We know that U^T creates all the colimits that $\mathcal{X}$ has and T preserves. Hence, U^T creates all coequalizers that $\mathcal{X}$ has of kernel pairs arising from $\mathcal{X}^T$, once the algebra structure is forgotten. Observe that e is such a coequalizer, in fact it is a regular epi and by *Proposition A.0.10* f and e share the same kernel pair, because of the factorization $f = me$ with m mono. Thus $(\text{Eq}(f), r_1, r_2)$ is also the kernel pair of e . Hence, by *Proposition A.0.6*, e is the coequalizer of $(\text{Eq}(f), r_1, r_2)$. In particular by the creation of such colimit we proved that Q has a structure of algebra, called ξ_Q , and that e can be lifted to a coequalizer in $\mathcal{X}^T$.

$$\begin{array}{ccccc}
 T(\text{Eq}(f)) & \xrightleftharpoons[T(r_2)]{T(r_1)} & T(A) & \xrightarrow{T(f)} & T(B) \\
 \downarrow \xi_{\text{Eq}(f)} & & \downarrow \xi_A & \searrow T(e) & \nearrow T(m) \\
 & & & T(Q) & \\
 & & & \downarrow \xi_Q & \\
 \text{Eq}(f) & \xrightleftharpoons[r_2]{r_1} & A & \xrightarrow{f} & B \\
 & & \searrow e & & \nearrow m \\
 & & & Q &
 \end{array}$$

In order to prove the factorization of f in $\mathcal{X}^T$, it remains to establish that m is a morphism in $\mathcal{X}^T$ and that it is a mono. Observe that m is a

mono in $\mathcal{X}^T$; in fact it is a mono in $\mathcal{X}$ and, furthermore, it is merely a morphism $\mathcal{X}$ that satisfies the algebra condition. We still need to establish that m is a morphism in $\mathcal{X}^T$.

$$\begin{aligned} f = me &\implies f\xi_A = me\xi_A \\ &\implies \xi_B T(f) = m\xi_Q T(e) \\ &\implies \xi_B T(m)T(e) = m\xi_Q T(e) \end{aligned}$$

We know that $T(e)$ is a regular epi, thus it is a epi. So we can cancel it from both sides and we obtain the condition for m to be a morphism in $\mathcal{X}^T$:

$$\xi_B T(m) = m\xi_Q$$

We proved that m and p are respectively a mono and a regular epi in $\mathcal{X}^T$. This allows the lifting of the factorization $f = me$ to $\mathcal{X}^T$.

It remains to prove the last condition of a regular category, that is the pullback stability of regular epi. Let $q : A \rightarrow B$ be a regular epi in $\mathcal{X}^T$, ie a coequalizer of some arrows in $\mathcal{X}^T$. In particular, q is a regular epi in $\mathcal{X}$. Since $\mathcal{X}$ is exact, q is pullback stable. Let $\tilde{q}$ be the pullback in $\mathcal{X}^T$ of q along some morphism. As just noticed, $\tilde{q}$ is a regular epi in $\mathcal{X}$. Let $(\text{Eq}(\tilde{q}), r_1, r_2)$ be the kernel pair of $\tilde{q}$. By *Proposition A.0.6*, $\tilde{q}$ is the coequalizer of such kernel pair. We want to lift $\tilde{q}$ to a regular epi in $\mathcal{X}^T$. By construction, the pullback considered is a pullback in $\mathcal{X}^T$, thus it comes with algebra structures of $A \times_B C$ and C . Let $\xi_{A \times_B C}$ and ξ_C be the algebra structures of $A \times_B C$ and C respectively. Moreover, $\tilde{q}$ is a morphism in $\mathcal{X}^T$. We know that the forgetful functor U^T creates all the limits that $\mathcal{X}$ has, so we can lift the kernel pair $(\text{Eq}(\tilde{q}), r_1, r_2)$ to a kernel pair in $\mathcal{X}^T$. Let $\xi_{\text{Eq}(\tilde{q})}$ be the algebra structure of $\text{Eq}(\tilde{q})$. Then we know that U^T creates all the colimits that $\mathcal{X}$ has and the monad T preserves, and we already proved that T preserves coequalizers of kernel pairs. Thus,

since $\tilde{q}$ is such a coequalizer, we can lift it to a coequalizer in $\mathcal{X}^T$.

$$\begin{array}{ccccccc}
 & & T(K) & \xRightarrow{\quad} & T(A) & \xrightarrow{T(q)} & T(B) \\
 & & \downarrow & & \downarrow & & \downarrow \\
 T(\text{Eq}(\tilde{q})) & \xRightarrow{T(r_1)} & & \nearrow & T(A \times_B C) & \xrightarrow{T(\tilde{q})} & T(C) \\
 & \searrow T(r_2) & & & \downarrow & & \downarrow \\
 & & K & \xRightarrow{\quad} & A & \xrightarrow{q} & B \\
 & & \downarrow \xi_{A \times_B C} & & \downarrow \xi_C & & \downarrow \\
 \text{Eq}(\tilde{q}) & \xRightarrow[r_2]{r_1} & A \times_B C & \xrightarrow{\quad} & C & \xrightarrow{\tilde{q}} & C
 \end{array}$$

□

Proposition 3.1.2. *Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a monad. If $\mathcal{X}$ is protomodular, then the Eilenberg-Moore category $\mathcal{X}^T$ is protomodular.*

Proof. Since the forgetful functor $\mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has, we can assert that $\mathcal{X}^T$ is finitely complete.

Then, let $f : (X, \xi) \rightarrow (Y, \xi')$ be a morphism in $\mathcal{X}^T$. The following diagram of functors is commutative. This essentially follows from the fact that the forgetful functor $\mathcal{X}^T \rightarrow \mathcal{X}$ creates all the limits that $\mathcal{X}$ has, in particular the pullbacks, which are used to define the change-of-base functor f^* .

$$\begin{array}{ccc}
 \text{Pt}_{\mathcal{X}^T}(Y, \xi') & \xrightarrow{f^*} & \text{Pt}_{\mathcal{X}^T}(X, \xi) \\
 \downarrow U^T & & \downarrow U^T \\
 \text{Pt}_{\mathcal{X}}(Y) & \xrightarrow{f^*} & \text{Pt}_{\mathcal{X}}(X)
 \end{array}$$

Let α be a morphism in $\text{Pt}_{\mathcal{X}^T}(Y, \xi')$. Suppose that $f^*(\alpha)$ is iso, we want to prove that α is iso. Since $f^*(\alpha)$ is iso, we have that $U^T(f^*(\alpha))$ is iso. By the commutativity of the diagram, we have that $U^T(f^*(\alpha)) = f^*(U^T(\alpha))$. Thus $f^*(U^T(\alpha))$ is an iso. Since $\mathcal{X}$ is protomodular, $U^T(\alpha)$ is an iso. Finally, U^T reflects isomorphisms, so α is an iso. □

3.2 Coproducts lift

We now proceed with results that will allow us to construct coproducts in the Eilenberg–Moore category, assuming they exist in the base category.

The construction of the coproducts in the Eilenberg–Moore category $\mathcal{X}^T$ will look quite intricate at first glance and it may appear overly complicated. However, this construction is needed because, given two objects with an algebra structure each, the coproduct of the two algebra structures is not an algebra structure in general. To make this clearer consider the following example.

Example 3.2.1. Let $\mathbf{Mon}$ be the category of monoids and let $U : \mathbf{Mon} \rightarrow \mathbf{Set}$ be the forgetful functor. One can define the free functor $F : \mathbf{Set} \rightarrow \mathbf{Mon}$ sending each set X to the free monoid on X

$$F(X) = \{(x_1, \dots, x_n) \mid n \in \mathbb{N}, \forall i = 1, \dots, n, x_i \in X\}$$

with the operation of concatenation. This induces a monad $T = UF$ on $\mathbf{Set}$ and the functor turns out to be monadic. The coproduct in $\mathbf{Mon}$ is the free product, while in $\mathbf{Set}$ it is the disjoint union. The free product of two monoids, as sets, is in general a bigger set than the disjoint union of the two sets underlying the monoids. Thus the forgetful functor U in general does not preserve coproducts. Instead, if the coproduct of the algebra structures, which correspond to the operation of the monoids, were itself an algebra structure, the forgetful functor would preserve coproducts.

$$\begin{array}{ccc} \mathbf{Mon} & \xrightarrow{\cong} & \mathbf{Set}^T \\ F \uparrow & \searrow U & \\ \mathbf{Set} & & \end{array}$$

The following will be rather intricate and technical; readers may choose to skip directly to the final outcome *Proposition 3.2.5*. The following results are based on the work of [Lin69].

For the sake of clarity, we first present a technical definition.

Definition 3.2.2. Let $U : \mathcal{X} \rightarrow \mathcal{A}$ be a functor, let X be an object in $\mathcal{X}$ and let $\{(f_i, g_i)\}_{i \in I}$ be a family of pair of parallel arrows in $\mathcal{A}$, where $f_i, g_i : A_i \rightarrow U(X)$. A morphism $q : X \rightarrow Q$ in $\mathcal{X}$ is said to be a ***U-coequalizer*** of the pair $\{(f_i, g_i)\}_{i \in I}$ if

- $U(q)f_i = U(q)g_i \forall i \in I$ in $\mathcal{A}$;

- If $h : X \rightarrow Y$ is a morphism in $\mathcal{X}$ such that $U(h)f_i = U(h)g_i \ \forall i \in I$ in $\mathcal{A}$, then there exists a unique morphism $t : Q \rightarrow Y$ in $\mathcal{X}$ such that $tq = h$.

$$\begin{array}{ccc}
 A & \xrightarrow[f_i]{g_i} & U(X) \xrightarrow{U(q)} U(Q) \\
 & & \searrow U(h) \quad \downarrow U(t) \\
 & & U(Y)
 \end{array}
 \qquad
 \begin{array}{ccc}
 X & \xrightarrow{q} & Q \\
 & \searrow h & \downarrow t \\
 & & Y
 \end{array}$$

Before continuing, we recall the universal property of the unit in an adjunction. Consider a functor $U : \mathcal{A} \rightarrow \mathcal{B}$ right adjoint to a functor F . Then for each object A in $\mathcal{A}$, B in $\mathcal{B}$, and $f : B \rightarrow U(A)$ in $\mathcal{B}$ there exists a unique morphism $g : F(B) \rightarrow A$ in $\mathcal{A}$ such that $f = U(g)\eta_B$. We denote $g = \bar{f}$.

$$\begin{array}{ccc}
 B & \xrightarrow{\eta_B} & UF(B) \\
 & \searrow f & \downarrow U(g) \\
 & & U(A)
 \end{array}
 \qquad
 \begin{array}{ccc}
 & & F(B) \\
 & & \downarrow g \\
 & & A
 \end{array}$$

Also recall that a monad (T, μ, η) induces a functor U^T and a functor F^T defined as follows:

$$\begin{array}{ccc}
 \mathcal{X}^T & & (X, \xi) \xrightarrow{U^T} X \\
 F^T \uparrow \downarrow U^T & & \\
 \mathcal{X} & & X \xrightarrow{F^T} (T(X), \mu_X)
 \end{array}$$

Lemma 3.2.3. *Let $U : \mathcal{X} \rightarrow \mathcal{A}$ be a functor right adjoint to a functor $F : \mathcal{A} \rightarrow \mathcal{X}$, let X be an object in $\mathcal{X}$ and let $\{(f_i, g_i)\}_{i \in I}$ be a family of parallel arrows in $\mathcal{X}$, where $f_i, g_i : A_i \rightarrow U(X)$. Consider the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ of unique morphisms induced by the universal property of the unit, such that $f_i = U(\bar{f}_i)\eta_{A_i}$ and $g_i = U(\bar{g}_i)\eta_{A_i}$. If $q : X \rightarrow Q$ is the coequalizer of the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ in $\mathcal{X}$, then q is the U -coequalizer of the family $\{(f_i, g_i)\}_{i \in I}$ in $\mathcal{A}$.*

$$\begin{array}{ccc}
 A_i & \xrightarrow{\eta_{A_i}} & UF(A_i) \\
 & \searrow f_i \quad \searrow g_i & \downarrow U(\bar{g}_i) \quad \downarrow U(\bar{f}_i) \\
 & & U(X)
 \end{array}
 \qquad
 \begin{array}{ccc}
 F(A_i) & \xrightarrow[\bar{g}_i]{\bar{f}_i} & X \xrightarrow{q} Q \\
 A_i \xrightarrow[g_i]{f_i} U(X) & \xrightarrow{U(q)} & U(Q)
 \end{array}$$

Proof. Assume $q : X \rightarrow Q$ is the coequalizer of the family $\{(\bar{f}_i, \bar{g}_i)\}_{i \in I}$ in $\mathcal{X}$, we want to prove that q is the U^T -coequalizer of the family $\{(f_i, g_i)\}_{i \in I}$ in $\mathcal{A}$. First we show, for each i , that $U(q)f_i = U(q)g_i$. Assume that $q\bar{f}_i = q\bar{g}_i$.

$$\begin{aligned} q\bar{f}_i = q\bar{g}_i &\implies U(q)U(\bar{f}_i) = U(q)U(\bar{g}_i) \\ &\implies U(q)U(\bar{f}_i)\eta_{A_i} = U(q)U(\bar{g}_i)\eta_{A_i} \\ &\implies U(q)f_i = U(q)g_i \end{aligned}$$

We now aim to prove the universal property of the U^T -coequalizer. Let $h : X \rightarrow Y$ be a morphism such that $U(h)f_i = U(h)g_i$. By the universal property of the unit applied to $U(h)f_i = U(h)g_i : A_i \rightarrow U(Y)$ there exist a unique morphism $k : F(A_i) \rightarrow Y$ such that $U(k)\eta_{A_i} = U(h)f_i = U(h)g_i$. Both hf_i and hg_i satisfies such condition; in fact $U(hf_i)\eta_{A_i} = U(h)U(\bar{f}_i)\eta_{A_i} = U(h)f_i$ and similarly for hg_i . Thus by the uniqueness $hf_i = k = hg_i$.

$$\begin{array}{ccc} A_i & \xrightarrow{\eta_{A_i}} & UF(A_i) \\ & \searrow f_i & \downarrow U(\bar{g}_i) \parallel U(\bar{f}_i) \\ & \searrow g_i & U(X) \\ & \searrow & \downarrow U(h) \\ & & U(Y) \end{array} \quad \begin{array}{ccc} F(A_i) & & \\ \bar{f}_i \downarrow \parallel \bar{g}_i & & \\ X & & \\ \downarrow h & & \\ Y & & \end{array}$$

$U(h)f_i = U(h)g_i$ (curved arrow from A_i to $U(Y)$) $U(k)$ (curved arrow from $UF(A_i)$ to $U(Y)$) k (curved arrow from $F(A_i)$ to Y)

Then, by the universal property of the coequalizer q in $\mathcal{X}^T$, there exists a unique morphism $t : Q \rightarrow Y$ such that $tq = h$. $\square$

Proposition 3.2.4. *Let $\mathcal{X}$ be a category, let (T, μ, η) be a monad on $\mathcal{X}$ and let $\{(A_i, \xi_i)\}_{i \in I}$ be a family of objects in $\mathcal{X}^T$. Assume that the coproducts $\coprod_i A_i$ and $\coprod_i T(A_i)$ exist in $\mathcal{X}$ with respective canonical injections $\{\iota_i\}_{i \in I}$ and $\{\iota'_i\}_{i \in I}$. If the following parallel arrows in $\mathcal{X}^T$ admits a coequalizer*

$$q : T(\amalg_i) \rightarrow (Q, \xi_Q)$$

$$\begin{array}{ccc}
 & (T^2(\amalg_i A_i), \mu) & \\
 T[\dots, T(\iota_i), \dots] \nearrow & & \searrow \mu_{\amalg_i A_i} \\
 (T(\amalg_i T(A_i)), \mu) & \xrightarrow{T(\amalg_i \xi_i)} & (T(\amalg_i A_i), \mu)
 \end{array} \quad (3.1)$$

then both the following diagrams of parallel arrow in $\mathcal{X}$ have q as U^T -coequalizer $\forall j \in I$.

$$\begin{array}{ccc}
 \amalg_i T(A_i) & \xrightarrow{[\dots, T(\iota_i), \dots]} & T(\amalg_i A_i) \\
 \searrow \amalg_i \xi_i & & \nearrow \eta_{\amalg_i A_i} \\
 & \amalg_i A_i &
 \end{array} \quad (3.2)$$

$$\begin{array}{ccc}
 T(A_j) & \xrightarrow{T(\iota_j)} & T(\amalg_i A_i) \\
 \searrow \xi_j & & \nearrow \eta_{\amalg_i A_i} \\
 & A_j \xrightarrow{\iota_j} \amalg_i A_i &
 \end{array} \quad (3.3)$$

Proof. Consider the diagram (3.1) of parallel arrows in $\mathcal{X}^T$, and p its coequalizer, here below explicitly.

$$\begin{array}{ccccc}
 & & T^3(\amalg_i A_i) & & \\
 & T^2[\dots, T(\iota_i), \dots] \nearrow & & \searrow T(\mu_{\amalg_i A_i}) & \\
 T^2(\amalg_i T(A_i)) & \xrightarrow{T(\amalg_i \xi_i)} & & & T^2(\amalg_i A_i) \\
 \downarrow \mu_{\amalg_i T(A_i)} & & \downarrow \mu_{T(\amalg_i A_i)} & & \downarrow \mu_{\amalg_i A_i} \\
 & T[\dots, T(\iota_i), \dots] \nearrow & T^2(\amalg_i A_i) & \searrow \mu_{\amalg_i A_i} & \\
 T(\amalg_i T(A_i)) & \xrightarrow{T(\amalg_i \xi_i)} & & & T(\amalg_i A_i)
 \end{array}$$

The top and bottom arrows from $(T(\amalg_i T(A_i)), \mu)$ to $(T(\amalg_i A_i), \mu)$ of the diagram (3.1) correspond, by the universal property of the unit of the

adjunction, respectively to the top and bottom arrows of the diagram (3.2). In order to prove so, by the uniqueness of the morphism induced by the universal property of the unit, it suffices to show that the image of the diagram (3.1) through U^T , composed with the unit, is the diagram (3.2). Such composition results to the following diagram.

$$\begin{array}{ccccc}
 & T(\coprod_i T(A_i)) & \xrightarrow{T[\dots, T(\iota_i), \dots]} & T^2(\coprod_i A_i) & \\
 \eta_{\coprod_i T(A_i)} \nearrow & & & & \searrow \mu_{\coprod_i A_i} \\
 \coprod_i T(A_i) & & (\spadesuit) & & T(\coprod_i A_i) \\
 \eta_{\coprod_i T(A_i)} \searrow & & & & \nearrow T(\coprod_i \xi_i) \\
 & T(\coprod_i T(A_i)) & & &
 \end{array}$$

Consider now the top arrow in the diagram $(\spadesuit)$ from $\coprod_i T(A_i)$ to $T(\coprod_i A_i)$. Using the naturality of η , we have that

$$\begin{array}{ccc}
 \coprod_i T(A_i) & \xrightarrow{\eta_{\coprod_i T(A_i)}} & T(\coprod_i T(A_i)) \\
 \downarrow [\dots, T(\iota_i), \dots] & & \downarrow T[\dots, T(\iota_i), \dots] \\
 T(\coprod_i A_i) & \xrightarrow{\eta_{T(\coprod_i A_i)}} & T^2(\coprod_i A_i)
 \end{array}$$

$$\begin{aligned}
 \mu_{\coprod_i A_i} T[\dots, T(\iota_i), \dots] \eta_{\coprod_i T(A_i)} &= \mu_{\coprod_i A_i} \eta_{T(\coprod_i A_i)} [\dots, T(\iota_i), \dots] \\
 &= 1_{T(\coprod_i A_i)} [\dots, T(\iota_i), \dots] \\
 &= [\dots, T(\iota_i), \dots]
 \end{aligned}$$

which is the top arrow in (3.2). Then, for the bottom arrow from $\coprod_i T(A_i)$ to $T(\coprod_i A_i)$ in $(\spadesuit)$, by the naturality of η and the property of the monad, we have that

$$\begin{array}{ccc}
 \coprod_i T(A_i) & \xrightarrow{\eta_{\coprod_i T(A_i)}} & T(\coprod_i T(A_i)) \\
 \downarrow \coprod_i \xi_i & & \downarrow T(\coprod_i \xi_i) \\
 \coprod_i A_i & \xrightarrow{\eta_{\coprod_i A_i}} & T(\coprod_i A_i)
 \end{array}$$

$$T(\coprod_i \xi_i) \eta_{\coprod_i T(A_i)} = \eta_{\coprod_i A_i} (\coprod_i \xi_i)$$

which is the bottom arrow in (3.2). By *Lemma 3.2.3*, it follows that the diagram (3.2) has p as U^T -coequalizer.

It remains to prove that the diagram (3.3) has p as U^T -coequalizer. We already know that (3.2) has p as U^T -coequalizer. Note that these two diagrams are related by the canonical injection $\iota'_j : T(A_j) \rightarrow \coprod_i T(A_i)$, as shown in the diagram below. In particular, by the universal property of the coproduct, it holds that $(\coprod_i \xi_i) \iota'_j = \iota_j \xi_j$ and $[\dots, T(\iota_i), \dots] \iota'_j = T(\iota_j)$.

$$\begin{array}{ccccc}
 & & T(\iota_j) & & \\
 & \nearrow & & \searrow & \\
 T(A_j) & \xrightarrow{\iota'_j} & \coprod_i T(A_i) & \xrightarrow{[\dots, T(\iota_i), \dots]} & T(\coprod_i A_i) \\
 & \searrow & \downarrow \coprod_i \xi_i & \nearrow \eta_{\coprod_i A_i} & \\
 & & \coprod_i A_i & &
 \end{array}$$

$\iota_j \xi_j$

The first condition of the U^T -coequalizer for (3.3) is obviously satisfied, just composing with ι'_j the condition of the U^T -coequalizer of (3.2):

$$\begin{aligned}
 U^T(q)[\dots, T(\iota_i), \dots] &= U^T(q) \eta_{\coprod_i A_i} \coprod_i \xi_i \implies \\
 \implies U^T(q)[\dots, T(\iota_i), \dots] \iota'_j &= U^T(q) \eta_{\coprod_i A_i} \coprod_i \xi_i \iota'_j \\
 \implies U^T(q) T(\iota_j) &= U^T(q) \iota_j \xi_j
 \end{aligned}$$

The universal property of the U^T -coequalizer of (3.3) also holds. In fact, let h be a morphism in $\mathcal{X}^T$ such that $U^T(h) T(\iota_j) = U^T(h) \iota_j \xi_j \ \forall j \in I$, then

$$\begin{aligned}
 U^T(h) T(\iota_j) &= U^T(h) \iota_j \xi_j \quad \forall j \in I \implies \\
 \implies U^T(h)[\dots, T(\iota_i), \dots] \iota'_j &= U^T(h) \eta_{\coprod_i A_i} \coprod_i \xi_i \iota'_j \quad \forall j \in I \\
 \implies U^T(h)[\dots, T(\iota_i), \dots] &= U^T(h) \eta_{\coprod_i A_i} \coprod_i \xi_i
 \end{aligned}$$

and by the universal property of the U^T -coequalizer of (3.2) there exists a unique morphism k such that $kq = h$. $\square$

The following gives a criteria to construct coproducts in the Eilenberg–Moore category, assuming they exist in the base category. This is possible if we assume the existence of reflexive coequalizers in the Eilenberg–Moore category.

Proposition 3.2.5. *Let $T : \mathcal{X} \rightarrow \mathcal{X}$ be a monad. If $\mathcal{X}$ has finite coproducts and the Eilenberg-Moore category $\mathcal{X}^T$ has reflexive coequalizers, then $\mathcal{X}^T$ has finite coproducts. Moreover $\mathcal{X}^T$ is finitely cocomplete.*

Proof. Let (T, μ, η) be the given monad, and suppose that $\mathcal{X}$ has finite coproducts. Let (A, ξ_A) and (B, ξ_B) be two objects in $\mathcal{X}^T$. We aim to prove that (A, ξ_A) and (B, ξ_B) have a coproduct in $\mathcal{X}^T$. All the coproducts in the diagram below are taken in $\mathcal{X}$, and therefore they exist. Let ι_1 and ι_2 be the canonical injections in $\mathcal{X}$ of A and B into $A + B$, respectively. Our goal is to prove that the diagram below is composed of a reflexive pair in $\mathcal{X}^T$. Then, by assumption, it admits a coequalizer in $\mathcal{X}^T$, denoted by q . This diagram correspond to the diagram (3.1) in *Proposition 3.2.4*.

$$\begin{array}{ccccc}
 & & T^3(A+B) & & \\
 & \nearrow T^2[T(\iota_1), T(\iota_2)] & \downarrow & \searrow T(\mu_{A+B}) & \\
 T^2(T(A) + T(B)) & \xleftarrow{T^2(\eta_A + \eta_B)} & T^2(A+B) & \xrightarrow{T(q)} & T(Q) \\
 & \xrightarrow{T^2(\xi_A + \xi_B)} & & & \\
 \downarrow \mu_{T(A)+T(B)} & & \downarrow \mu_{T(A+B)} & & \downarrow \xi_Q \\
 & \nearrow T[T(\iota_1), T(\iota_2)] & T^2(A+B) & \searrow \mu_{A+B} & \\
 T(T(A) + T(B)) & \xleftarrow{T(\eta_A + \eta_B)} & T(A+B) & \xrightarrow{q} & Q \\
 & \xrightarrow{T(\xi_A + \xi_B)} & & &
 \end{array}$$

The parallel arrows in $\mathcal{X}^T$ to be coequalized are

- $T(\xi_A + \xi_B)$
- $\mu_{A+B}T[T(\iota_1), T(\iota_2)]$

We proceed to show that they form a reflexive pair by exhibiting a common section. $T(\eta_A + \eta_B)$ is such a common section, as shown below. It follows the first composition

$$T[\xi_A, \xi_B]T(\eta_A + \eta_B) = T([\xi_A, \xi_B](\eta_A + \eta_B)) = T(1_{A+B}) = 1_{T(A+B)}$$

and the second composition, using the naturality of η .

$$\begin{aligned}
\mu_{A+B}T[T(\iota_1), T(\iota_2)]T(\eta_A + \eta_B) &= \mu_{A+B}T([T(\iota_1), T(\iota_2)](\eta_A + \eta_B)) \\
&= \mu_{A+B}T[T(\iota_1)\eta_A, T(\iota_2)\eta_B] \\
&= \mu_{A+B}T[\eta_{A+B}\iota_1, \eta_{A+B}\iota_2] \\
&= \mu_{A+B}T(\eta_{A+B}[\iota_1, \iota_2]) \\
&= \mu_{A+B}T(\eta_{A+B}1_{A+B}) \\
&= \mu_{A+B}T(\eta_{A+B}) = 1_{T(A+B)}
\end{aligned}$$

Thus it is a reflexive pair.

Let $q : T(A + B) \rightarrow Q$ be the coequalizer in $\mathcal{X}$, and ξ_Q be the algebra structure of Q . We aim to prove that (Q, ξ_Q) is the coproduct of (A, ξ_A) and (B, ξ_B) in $\mathcal{X}^T$. First we need to exhibit the canonical injections $\iota_{Q,1}$ and $\iota_{Q,2}$. They are respectively defined as the following compositions.

$$A \xrightarrow{\iota_1} A + B \xrightarrow{\eta_{A+B}} T(A + B) \xrightarrow{q} Q$$

$$B \xrightarrow{\iota_2} A + B \xrightarrow{\eta_{A+B}} T(A + B) \xrightarrow{q} Q$$

We still need to prove that they are morphisms^a in $\mathcal{X}^T$. We prove the first one, the second one is similar. Consider the following diagram. All diagrams, except the left and the external ones, commute. We aim to prove that the left diagram commutes, so that the external one commutes as well. This holds by *Proposition 3.2.4*, since it corresponds to the diagram (3.3).

$$\begin{array}{ccccccc}
T(A) & \xrightarrow{T(\iota_1)} & T(A + B) & \xrightarrow{T(\eta_{A+B})} & T^2(A + B) & \xrightarrow{T(q)} & T(Q) \\
\downarrow \xi_A & & & \searrow & \downarrow \mu_{A+B} & & \downarrow \xi_Q \\
& & & & T(A + B) & & \\
& & & & \searrow q & & \\
A & \xrightarrow{\iota_1} & A + B & \xrightarrow{\eta_{A+B}} & T(A + B) & \xrightarrow{q} & Q
\end{array}$$

Once we have the canonical injections, we need to show that they satisfy the universal property of coproducts in $\mathcal{X}^T$. Let $f_i : (A_i, \xi_i) \rightarrow (K, \xi_K)$ for $i = 1, 2$ in $\mathcal{X}^T$. We aim to prove that there exists a unique morphism $h : (Q, \xi_Q) \rightarrow (K, \xi_K)$ in $\mathcal{X}^T$ such that $h\iota_{Q,i} = f_i$ for $i = 1, 2$. Refer to

the diagram at the end of the proof for a visualization of the following arguments.

First, we show that there exists a unique morphism $g : (T(A + B), \mu_{A+B}) \rightarrow (K, \xi_K)$ in $\mathcal{X}^T$ such that $g(\eta_{A+B})\iota_i = f_i$ for $i = 1, 2$. Define $g = \xi_K T[f_1, f_2]$. It is a morphism in $\mathcal{X}^T$, in fact the following diagram commutes.

$$\begin{array}{ccccc}
 T^2(A + B) & \xrightarrow{T[f_1, f_2]} & T^2(K) & \xrightarrow{\mu_K} & T(K) \\
 \downarrow \mu_{A+B} & & \downarrow \mu_K & & \downarrow \xi_K \\
 T(A + B) & \xrightarrow{T[f_1, f_2]} & T(K) & \xrightarrow{\xi_K} & K
 \end{array}$$

Suppose now that there exists an other morphism $g' : (T(A + B), \mu_{A+B}) \rightarrow (K, \xi_K)$ such that $g'(\eta_{A+B})\iota_i = f_i$ for $i = 1, 2$. Then we have that

$$\begin{aligned}
 g(\eta_{A+B})\iota_i &= g'(\eta_{A+B})\iota_i \quad i = 1, 2 \\
 \implies g(\eta_{A+B}) &= g'(\eta_{A+B}) \\
 \implies U^T(g)(\eta_{A+B}) &= U^T(g')(\eta_{A+B}) \\
 \implies g &= g'
 \end{aligned}$$

where we used the uniqueness in the universal property of the unit and the fact that g and g' are morphisms in $\mathcal{X}^T$. So g is unique. Moreover, we have that $gT(\iota_i) = g(\eta_{A+B})\iota_i \xi_i$ for $i = 1, 2$. In fact, we have that

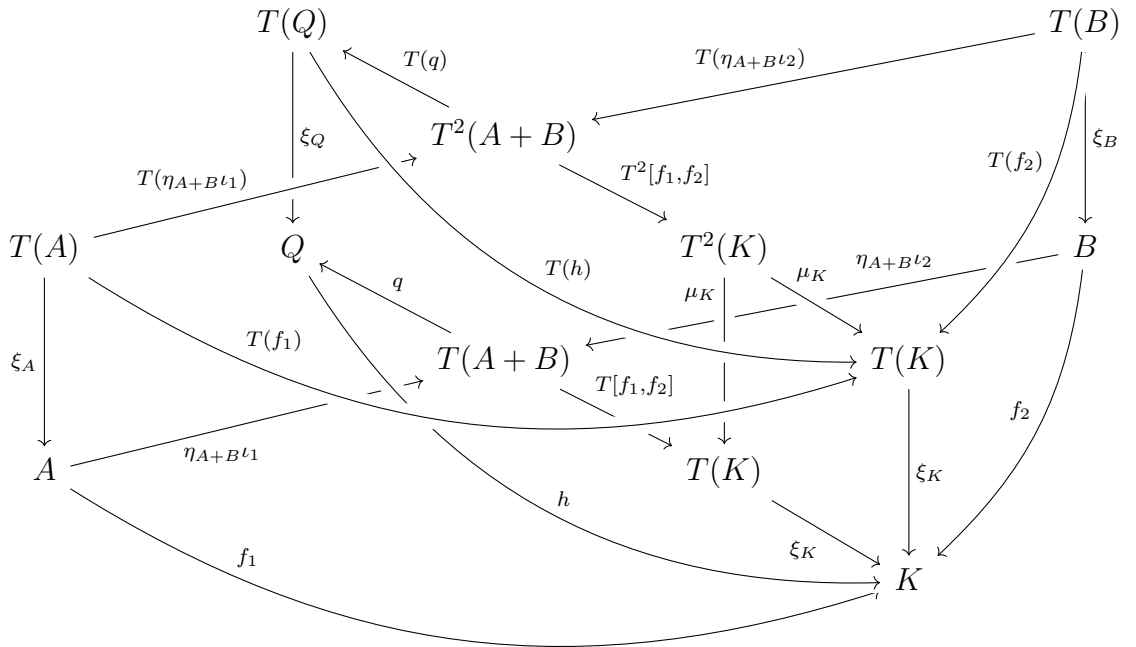
$$\begin{array}{ccccccc}
 & & T(A + B) & & & & \\
 & \nearrow T(\iota_1) & & \searrow T[f_1, f_2] & & & \\
 T(A) & \xrightarrow{T(f_1)} & T(K) & \xrightarrow{\xi_K} & K & & \\
 \downarrow \xi_A & & \nearrow f_1 & & \uparrow \xi_K & & \\
 & A & \xleftarrow{\xi_A} & T(A) & & & \\
 & \nearrow \eta_A & & \downarrow T(\iota_1) & \searrow T(f_1) & & \\
 A & \xrightarrow{\iota_1} & A + B & \xrightarrow{\eta_{A+B}} & T(A + B) & \xrightarrow{T[f_1, f_2]} & T(K)
 \end{array}$$

$$\begin{aligned}
gT(\iota_1) &= \xi_K T[f_1, f_2] T(\iota_i) = \xi_K T([f_1, f_2] \iota_i) \\
&= \xi_K T(f_1) = f_1 \xi_A = f_1 1_A \xi_A \\
&= f_1 \xi_A \eta_A \xi_A = \xi_K T(f_1) \eta_A \xi_A \\
&= \xi_K T[f_1, f_2] T(\iota_1) \eta_A \xi_A \\
&= \xi_K T[f_1, f_2] (\eta_{A+B}) \iota_1 \xi_A \\
&= g(\eta_{A+B}) \iota_1 \xi_A
\end{aligned}$$

and the same holds for B . Therefore g has equal compositions with the parallel arrows corresponding to (3.3), which has, by *Proposition 3.2.4*, q as U^T -coequalizer. Then by the universal property of the U^T -coequalizer, it follows that there exists a unique morphism $h : (Q, \xi_Q) \rightarrow (K, \xi_K)$ such that $g = hq$ in $\mathcal{X}^T$. Thus, we have that

$$h\iota_{Q,i} = hq(\eta_{A+B})\iota_i = g(\eta_{A+B})\iota_i = f_i \quad i = 1, 2$$

and by the uniqueness of g , we have that h is the unique morphism in $\mathcal{X}^T$ satisfying such relation.



In order to prove that also finite coproducts exists, it suffices to do the same proof for the initial object, using *Proposition 3.2.4* again.

It remains to prove that $\mathcal{X}^T$ is finitely cocomplete. We already know that it has finite coproducts, so we only need to show that it has coequalizers. This follows from the fact that it has finite coproducts and reflexive coequalizers; in fact each coequalizer can be constructed as a reflexive coequalizer from the coproduct, as shown below.

$$\begin{array}{ccc} A & \xrightleftharpoons[g]{f} & B \\ A + B & \xrightleftharpoons[g, 1_B]{[f, 1_B]} & B \end{array}$$

□

^aNote that ι_1 and η_{A+B} are not morphisms in $\mathcal{X}^T$. This is not a problem, however, since we will show that the whole composition $q\eta_{A+B}\iota_1$ is a morphism in $\mathcal{X}^T$

We now want to study a sufficient condition for a category to admit reflexive coequalizers, so that we can apply the proposition above.

Recall that a pair (f, g) from A to B in a regular category induces a relation with the following construction. Consider the morphism induced by the universal property $\langle f, g \rangle$. Since the category is regular, we can take the factorization $\langle f, g \rangle = \langle r_1, r_2 \rangle e$, with $\langle r_1, r_2 \rangle$ mono and e regular epi. Then $\langle r_1, r_2 \rangle$ is the relation induced.

$$\begin{array}{ccc} A & \xrightarrow{\langle f, g \rangle} & B \times B \\ & \searrow e & \nearrow \langle r_1, r_2 \rangle \\ & R & \end{array}$$

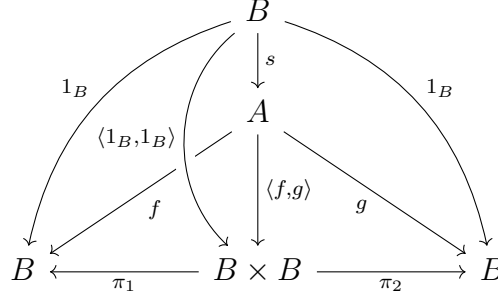
Proposition 3.2.6. *The internal relation induced by a reflexive pair (f, g) in a regular category is reflexive.*

Proof. Consider the pair (f, g) from A to B and the section s with $fs = 1_B$ and $gs = 1_B$.

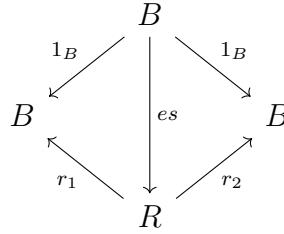
$$A \xrightleftharpoons[g]{f} B$$

Let $\langle f, g \rangle = \langle r_1, r_2 \rangle e$ be the (regular epi, mono) factorization. Moreover,

by the universal property we have that $\langle f, g \rangle s = \langle 1_B, 1_B \rangle$.



Thus $\langle r_1, r_2 \rangle es = \langle 1_B, 1_B \rangle$. The morphism es makes the relation reflexive.



□

Proposition 3.2.7. *Let $\mathcal{A}$ be an exact Mal'tsev category. Then $\mathcal{A}$ has reflexive coequalizers.*

Proof. Assume $\mathcal{A}$ is exact and Mal'tsev. We want to prove that $\mathcal{A}$ has reflexive coequalizer. Consider a reflexive pair (f, g) from A to B . By *Proposition 3.2.6* the internal relation induced $\langle r_1, r_2 \rangle$ is reflexive. Since $\mathcal{A}$ is Mal'tsev, then each internal reflexive relation is an equivalence relation. Hence $\langle r_1, r_2 \rangle$ is an equivalence relation. $\mathcal{A}$ is also exact, then, by *Proposition A.0.9* the equivalence relation (r_1, r_2) has a coequalizer k .

We want to prove that k is also the coequalizer of (f, g) . First we note that $kf = kr_1e = kr_2e = kg$. It remains to prove the universal property. Let $h : B \rightarrow C$ be a morphism such that $hf = hg$. Then $hr_1e = hr_2e$. Since e is a regular epi, hence by *Proposition A.0.8* an epi, we have that

$hr_1 = hr_2$. Thus there exists a unique morphism t such that $th = k$.

$$\begin{array}{ccccc}
 A & & & & \\
 \downarrow e & \searrow f & & & \\
 & g & & & C \\
 & \searrow & \nearrow h & & \downarrow \exists! t \\
 R & \xrightleftharpoons[r_2]{r_1} B & \xrightarrow{k} & \text{Coeq}(r_1, r_2) &
 \end{array}$$

□

Chapter 4

Effective descent morphisms

In this chapter we prove a monadicity theorem for the pullback functor. The standard terminology in the literature is *descent morphism* for a morphism whose pullback functor is premonadic, and *effective descent morphism* for a morphism whose pullback functor is monadic. We do not aim to develop descent theory; therefore, we will stick to the monadic theory nomenclature.

This theorem is usually stated in the literature in the descent theory framework, and then by a comparison theorem it holds in the monadic theory. The reader can refer to [JST03] for the first approach. We aim to prove this monadicity theorem directly in monadic theory, without the need of the comparison theorem. This part is based on the work of Sobral and Tholen [ST92], but we will introduce some changes to suit our context. Some ideas drawn from Janelidze and Tholen [JT91]. Let us first state the theorem.

(IMPLEMENT NOTES)

Chapter 5

Ideally exact categories

5.1 Definition and characterization

We are now ready to prove the theorem, which will lead to the definition of the ideally exact categories. The following characterization and subsequent results are based on the work of Janelidze [Jan24].

Theorem 5.1.1. *Let $\mathcal{A}$ be a category. The following are equivalent:*

1. $\mathcal{A}$ is exact and protomodular, has finite coproducts and the morphism $! : 0 \rightarrow 1$ is a regular epimorphism.
2. $\mathcal{A}$ is exact, has finite coproducts and there exists $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$.

$$\begin{array}{ccc}
 \mathcal{A} & \xrightarrow{\cong} & \mathcal{X}^{UF} \\
 \uparrow F & & \nearrow \\
 \mathcal{X} & \xleftarrow{U} &
 \end{array}$$

3. There exist $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ and F left adjoint to U such that the associated monad $T = UF$ preserves regular epimorphisms and kernel pairs.

Definition 5.1.2. *A category $\mathcal{A}$ is said to be **ideally exact** if it satisfies one of the equivalent conditions of Theorem 5.1.1.*

Proof of Theorem 5.1.1. (1 $\implies$ 3) Suppose $! : 0 \rightarrow 1$ is a regular epi. Since $\mathcal{A}$ is exact, the pullback functor $(!)^* : (\mathcal{A} \downarrow 1) \rightarrow (\mathcal{A} \downarrow 0)$ is monadic by (SECTION TO DO). Note that $(\mathcal{A} \downarrow 1) \cong \mathcal{A}$. Now consider $\mathcal{X} = (\mathcal{A} \downarrow 0)$, we want to prove that $\mathcal{X}$ is semiabelian.

$(\mathcal{A} \downarrow 0)$ is pointed, in fact if (A, α) is an object in $(\mathcal{A} \downarrow 0)$, we have the following initial and terminal objects, which coincide:

$$\begin{array}{ccc} 0 & \xrightarrow{!_A} & A \\ & \searrow & \downarrow \alpha \\ & & 0 \end{array} \qquad \begin{array}{ccc} A & \xrightarrow{\alpha} & 0 \\ \downarrow \alpha & \swarrow 1_0 & \\ 0 & & \end{array}$$

$\mathcal{A}$ has finite coproducts, so $(\mathcal{A} \downarrow 0)$ has finite coproducts. In fact consider $(A, \alpha), (B, \beta)$ in $(\mathcal{A} \downarrow 0)$ and $A + B$ in $\mathcal{A}$, with the canonical injections $\iota_1 : A \rightarrow A + B$ and $\iota_2 : B \rightarrow A + B$. Let (C, γ) be an object in $(\mathcal{A} \downarrow 0)$ with $f : A \rightarrow C$ and $g : B \rightarrow C$ morphisms in $(\mathcal{A} \downarrow 0)$. Providing to $A + B$ the morphism $[\alpha, \beta]$ we have that $(A + B, [\alpha, \beta])$ is the coproduct of (A, α) and (B, β) in $(\mathcal{A} \downarrow 0)$, in fact $\gamma[f, g] = [\alpha, \beta]$, so $[f, g]$ is also a morphism in $(\mathcal{A} \downarrow 0)$.

$$\begin{array}{ccccc} A & \xrightarrow{\iota_1} & A + B & \xleftarrow{\iota_2} & B \\ & \searrow \alpha & \downarrow [\alpha, \beta] & \swarrow \beta & \\ & & 0 & & \\ & \searrow f & \uparrow \gamma & \swarrow g & \\ & & C & & \end{array}$$

Now we want to prove that $(\mathcal{A} \downarrow 0)$ is exact, from the exactness of $\mathcal{A}$. First we prove it is finitely complete. It suffices to show that it has terminal a object and pullbacks. The terminal object is the one just shown above. It also has pullbacks, by *Proposition 2.1.4*, since pullbacks are connected limits and $\mathcal{A}$ has them.

Consider now an equivalence relation (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$, endowed with the morphism ρ and α as shown in the diagram below. We want to prove that such equivalence relation is the kernel pair of some arrow. Since $\mathcal{A}$ is exact, it is a kernel pair in $\mathcal{A}$. By *Proposition A.0.9* we can suppose that it is the kernel pair of its coequalizer.

$$\begin{array}{ccccc} R & \xrightarrow{r_1} & A & \xrightarrow{q} & B \\ & \searrow r_2 & \downarrow \alpha & \swarrow \exists! \beta & \\ & \searrow \rho & 0 & & \end{array}$$

Since r_1 and r_2 are morphism in $(\mathcal{A} \downarrow 0)$, we have that $\alpha r_1 = \rho = \alpha r_2$. Thus, by the universal property of the coequalizer, there exists a unique morphism $\beta : B \rightarrow 0$ such that $\beta q = \alpha$. Providing to B the morphism β , we have that (B, β) is the coequalizer of (R, r_1, r_2) in $(\mathcal{A} \downarrow 0)$.

We proceed by proving that $(\mathcal{A} \downarrow 0)$ is regular. Both the conditions of the definition of regular category are easily extendable from $\mathcal{A}$ to $(\mathcal{A} \downarrow 0)$. In fact since in $\mathcal{A}$ each morphism admits a (regular epi, mono) factorization, then the same holds in $(\mathcal{A} \downarrow 0)$. Let $f = me$ be such a factorization through Q , with f morphism in $(\mathcal{A} \downarrow 0)$. Endow Q with the morphism t induced by the universal property of the coequalizer. Now we want to prove that e and m are morphism in $(\mathcal{A} \downarrow 0)$. Then they will be respectively regular epi and mono in $(\mathcal{A} \downarrow 0)$. Notice first that e is obviously a morphism in $(\mathcal{A} \downarrow 0)$, by the condition of t . Moreover, we have that $te = \alpha = \beta f = \beta me$ and since e is a regular epi, hence by *Proposition A.0.8* an epi, it follows that m is a morphism in $(\mathcal{A} \downarrow 0)$.

$$\begin{array}{ccccc}
 & & Q & & \\
 & \nearrow e & \vdots & \nwarrow m & \\
 A & \xrightarrow{f} & & \xrightarrow{\quad} & B \\
 & \searrow \alpha & \vdots \exists! t & \swarrow \beta & \\
 & & 0 & &
 \end{array}$$

In order to prove that regular epis are pullback stable in $(\mathcal{A} \downarrow 0)$ we can use the fact that they are in $\mathcal{A}$. In fact, we just endow the regular epi with the morphism induced by universal property of the coequalizer.

We must now establish that the $(\mathcal{A} \downarrow 0)$ is protomodular. Since the construction of the change-of-base functor involves pullbacks, which are connected limits, then by *Remark 2.1.3* the construction is the same in $(\mathcal{A} \downarrow 0)$ and in $\mathcal{A}$. Thus $(\mathcal{A} \downarrow 0)$ inherits protomodularity from $\mathcal{A}$, noting that the definition of iso in $\mathcal{A}$ and $(\mathcal{A} \downarrow 0)$ is essentially the same, except the morphism condition. Hence, since $\mathcal{X} = (\mathcal{A} \downarrow 0)$ is exact, protomodular, pointed and has binary coproducts, we can conclude that it is semiabelian.

It remains to prove that the monad $(!)*(!)_!$ preserves regular epimorphisms and kernel pairs. Consider a regular epi $q : A \rightarrow B$ in $(\mathcal{A} \downarrow 0)$, it is the coequalizer of some parallel arrows. Since the coequalizer is a colimit, it is preserved by the left adjoint functor $(!)_!$. So it suffices to prove that,

with $\mathcal{A}$ regular, $(!)^*$ preserves regular epis. Consider the following diagram:

$$\begin{array}{ccc}
 0 \times_1 A & \xrightarrow{(!)^*(q)} & 0 \times_1 B \\
 \downarrow & & \downarrow \\
 A & \xrightarrow{q} & B \\
 & \searrow & \swarrow \\
 & 1 &
 \end{array}$$

By *Property 2.2.4* the square is a pullback. Since $\mathcal{A}$ is regular, and hence so is $(\mathcal{A} \downarrow 0)$, it follows that $(!)^*(q)$ is a regular epi due to the stability of regular epis under pullbacks.

Finally, we want to establish that the monad $(!)^*(!)_!$ preserves kernel pairs. This is obvious: $(!)_!$ preserves connected limits, such as kernel pairs, by *Proposition 2.2.1*, while $(!)^*$ preserves kernel pairs since it is a right adjoint. Thus $(!)^*(!)_!$ preserves kernel pairs.

(3 $\implies$ 2) Suppose there exists $\mathcal{X}$ semiabelian category and a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$. Assume that the associated monad T preserves regular epimorphisms and kernel pairs.

First we need to verify that $\mathcal{A}$ is exact. Since U is monadic, we have that $\mathcal{A} \cong \mathcal{X}^T$. So it suffices to show that $\mathcal{X}^T$ is exact. We know that $\mathcal{X}$ is semiabelian, so it is exact. Then, by *Proposition 3.1.1*, $\mathcal{X}^T$ is exact.

In order to prove that $\mathcal{A}$ has finite coproducts, it is first necessary to establish that $\mathcal{X}^T$ is protomodular. This follows directly from *Proposition 3.1.2* as $\mathcal{X}$ is protomodular. Since any protomodular category is Mal'tsev, then $\mathcal{X}^T$ is Mal'tsev. We already prove that $\mathcal{X}^T$ is also exact, then by *Proposition 3.2.7*, $\mathcal{X}^T$ has reflexive coequalizers. Since $\mathcal{X}$ has finite coproducts and $\mathcal{X}^T$ has reflexive coequalizers, then by *Proposition 3.2.5*, $\mathcal{X}^T$ has finite coproducts. Finally, since $\mathcal{A} \cong \mathcal{X}^T$, it follows that $\mathcal{A}$ has finite coproducts.

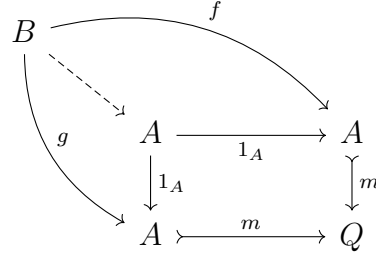
(2 $\implies$ 1) Suppose that $\mathcal{A}$ is exact, has finite coproducts and there exists a semiabelian category $\mathcal{X}$ and a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$.

Since $\mathcal{X}$ is protomodular, so is $\mathcal{X}^T$ by *Proposition 3.1.2*. Again, by monadicity, $\mathcal{A}$ inherits this properties.

We are left to show that the morphism $! : 0 \rightarrow 1$ is a regular epi. Since $\mathcal{A}$ is exact by assumption, the morphism $!$ admits a (regular epi, mono) factorization $! = me$.

Now we aim to prove that if a morphism m is a mono, then $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$, ie if

$f, g : U(A) \rightarrow U(B)$ for some A, B in $\mathcal{A}$, then $mf = mg \implies f = g$ ^a. First, we prove that if m is a mono, then $U^T(m)$ is a mono. Let m be a mono and A its domain. Then $(A, 1_A, 1_A)$ is the kernel pair of m , in fact, given f and g morphisms such that $mf = mg$, we obtain the universal property by the mono.



Since U^T is a right adjoint, it preserves limit, such as kernel pairs. Thus also the following diagram is a kernel pair.

$$\begin{array}{ccc}
 U^T(A) & \xrightarrow{1_{U^T(A)}} & U^T(A) \\
 \downarrow 1_{U^T(A)} & & \downarrow U(m) \\
 U^T(A) & \xrightarrow{U(m)} & Q
 \end{array}$$

This implies that $U^T(m)$ is a mono by the universal property of the kernel pair.

It remains to prove that the image of a mono through the comparison functor $K : \mathcal{A} \rightarrow \mathcal{X}^T$ acts as a mono in the way described above. Then, since $U = U^T K$, we have that U does the same. By monadicity, the functor K is fully faithful, this is sufficient to prove that K preserves monos. In fact, let m be a mono in $\mathcal{A}$ and let f, g be morphisms in $\mathcal{X}^T$ between object coming from $\mathcal{A}$ such that $K(m)f = K(m)g$. Let f' and g' be the morphisms in $\mathcal{A}$ such that $K(f') = f$ and $K(g') = g$ by the fully

property of K . Then we have that

$$\begin{aligned}
K(m)f &= K(m)g \\
\implies K(m)K(f') &= K(m)K(g') \\
\implies K(mf') &= K(mg') \\
\implies mf' &= mg' \\
\implies f' &= g' \\
\implies K(f') &= K(g') \\
\implies f &= g
\end{aligned}$$

Thus $K(m)$ acts as a mono in the morphisms in $\mathcal{X}^T$ between object coming from $\mathcal{A}$. By the above discussion, $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$.

Picking up where we left off, consider m the mono arising from the (regular epi, mono) factorization of $! : 0 \rightarrow 1$ in $\mathcal{A}$. By the discussion above, $U(m)$ acts as a mono in the morphisms in $\mathcal{X}$ between object coming from $\mathcal{A}$. In particular, since the final object 1 is a limit, it is preserved by U , then $U(m)$ acts as mono and has codomain 1 in $\mathcal{X}$. Moreover, $\mathcal{X}$ is semiabelian, hence pointed, so $0 = 1$ in $\mathcal{X}$. This means that $U(m)$ acts as a mono and has codomain 0 in $\mathcal{X}$. Finally, this implies that $U(m)$ is an iso, in fact, suppose that $U(m)$ acts as a mono with codomain 0 and let $U(C)$ be its domain. Denote by $!_{U(C)}$ the unique morphism from $U(C)$ to 0 , then $!_{U(C)}$ is the inverse of $U(m)$. In fact $!_{U(C)}U(m) = !_0 = 1_0$. Then composing with $U(m)$ we obtain $U(m)!_{U(C)}U(m) = U(m)$, thus, observing that $U(m)!_{U(C)}$ and $1_{U(C)}$ are a morphisms between objects arising from $\mathcal{A}$, we also have that $U(m)!_{U(C)} = 1_{U(C)}$. Thus $U(m)$ is an iso.

$$U(C) \xrightarrow{U(m)} 0 \quad 0 \xrightarrow{!_{U(C)}} U(C)$$

Now we want to prove that U reflects iso. As done before we prove that for the comparison functor K and for U^T , then it follows for U . K reflect iso because, by monadicity, it is fully faithful. In fact, if $K(f) : K(X) \rightarrow K(Y)$ is an iso and $g : K(Y) \rightarrow K(X)$ is its inverse in $\mathcal{X}^T$, then $K(f)g = 1_{K(Y)}$ and $gK(f) = 1_{K(X)}$. Let g' be the morphism in $\mathcal{A}$ such that $K(g') = g$ by

the fully property of K . So we have that

$$\begin{aligned}
 1_{K(Y)} &= K(f)g \\
 \implies 1_{K(Y)} &= K(f)K(g') \\
 \implies K(1_Y) &= K(f)K(g') \\
 \implies K(1_Y) &= K(fg') \\
 \implies 1_Y &= fg'
 \end{aligned}$$

and similarly $1_X = g'f$. Thus f is an iso in $\mathcal{A}$.

Also U^T reflects iso. In fact, let $U^T(k) : U^T(V) \rightarrow U^T(W)$ be an iso in $\mathcal{X}^T$ and $h : U^T(W) \rightarrow U^T(V)$ its inverse. We aim to prove that h is also a morphism in $\mathcal{X}^T$, so that it is an inverse for k in $\mathcal{X}^T$. Let ξ_V and ξ_W be respectively the algebra structures of $U^T(V)$ and $U^T(W)$. Then, since k is a morphism in $\mathcal{X}^T$, we have that $\xi_W T(k) = k\xi_V$, denoting $U^T(k)$ just as k . Composing with h and $T(h)$ we obtain

$$\begin{array}{ccc}
 T(V) & \xrightleftharpoons[T(h)]{T(k)} & T(W) \\
 \downarrow & & \downarrow \\
 V & \xrightleftharpoons[T(h)]{k} & W
 \end{array}$$

$$\begin{aligned}
 \xi_W T(k) &= k\xi_V \\
 \implies h\xi_W T(k)T(h) &= hk\xi_V T(h) \\
 \implies h\xi_W &= \xi_V T(h)
 \end{aligned}$$

Thus h is a morphism in $\mathcal{X}^T$, and k is an iso in $\mathcal{X}^T$. This implies, as discussed above, that also U reflects iso.

Finally we want to prove that $! : 0 \rightarrow 1$ is a regular epi in $\mathcal{A}$. We already know that $U(m)$ is an iso. Since U reflects iso, then m is an iso. Thus $! = me$ is a regular epi.

This holds because an iso, as m , is also a strong epimorphism, and e is a regular epimorphism, hence it is also a strong epimorphism; therefore,

the composition me is a strong epimorphism. Since the $\mathcal{A}$ is regular, then by a standard result $! = me$ is a regular epi, as claimed. $\square$

^aThis will be crucial to utilize the fully functor property of the comparison functor K .

Proposition 5.1.3. *An ideally exact category is cocomplete.*

Proof. An ideally exact category is exact and protomodular, thus Mal'tsev. So by *Proposition 3.2.7* it has reflexive coequalizers. Then *Proposition 3.2.5* implies that the category has all the finite colimits. $\square$

Proposition 5.1.4. *Let $\mathcal{A}$ be an ideally exact category. There exist $\mathcal{X}$ semiabelian category with a monadic functor $U : \mathcal{A} \rightarrow \mathcal{X}$ and the corresponding monad is essentially nullary.*

Proof. We can consider $(!)^*$ to be such functor, as shown in the proof of *Theorem 5.1.1* ($1 \implies 3$). Then by *Theorem 2.3.4* we have that the monad associated $(!)^*(!)_!$ is essentially nullary. $\square$

5.2 Main theorem

Appendix

Some of the following results are due to [BB04].

Definition A.0.1. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$. We say that f is a **strong epimorphism** if it for each commutative diagram of solid arrows

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ u \downarrow & \exists! d \swarrow & \downarrow v \\ C & \xrightarrow{m} & D \end{array}$$

where m is a mono, there exists a unique morphism d such that $df = u$ and $md = v$.

We say that f is an **extremal epimorphism** if given a factorizations $f = mf'$, with m mono, then m is an iso.

We say that f is a **regular epimorphism** if it is the coequalizer of some parallel arrows.

Proposition A.0.2. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$. The following are equivalent:

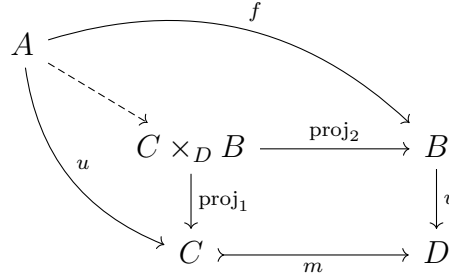
1. f is a strong epimorphism;
2. f is an extremal epimorphisms.

Proof. (1 $\implies$ 2) We want to prove that f is an extremal epi. Let $f = mg$ be a factorization with m mono and consider the square:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ g \downarrow & d \swarrow & \downarrow 1_B \\ C & \xrightarrow{m} & B \end{array}$$

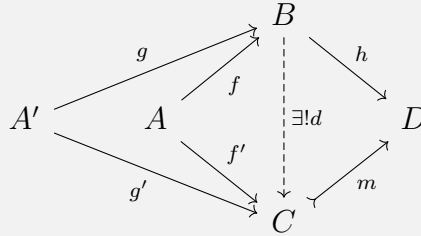
By (1) we have $md = 1_B$. Furthermore $mdm = m$ and, since m is mono, we have that $dm = 1_C$. Thus m is an iso.

(2 $\implies$ 1) We want to prove that f is a strong epi. Consider the pullback



Since m is a mono, the projection proj_2 is a mono. Now f factors through proj_2 , so by (2) proj_2 is an iso. We define $d = \text{proj}_1(\text{proj}_2)^{-1}$, which is the morphism we are looking for. $\square$

Definition A.0.3. Let $\mathcal{C}$ be a category and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. We say that the pair (f, g) is a **strongly epimorphic pair** if for each commutative diagram of solid arrows



where m is a mono, there exists a unique morphism d such that $f' = df$, $g' = dg$ and $md = h$.

We say that the pair (f, g) is an **extremally epimorphic pair** if given factorizations $f = mf'$ and $g = mg'$, with m mono, then m is an iso.

Proposition A.0.4. Let $\mathcal{C}$ be a category with pullbacks and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:

1. (f, g) is a strongly epimorphic pair;
2. (f, g) is a extremally epimorphic pair;

Proof. By adding the morphism g' and its conditions in the proof of Proposition A.0.2, we obtain the proof of the current proposition. $\square$

Proposition A.0.5. *Let $\mathcal{C}$ be a category with binary coproducts and $f : A \rightarrow B$, $g : A' \rightarrow B$ be morphisms. The following are equivalent:*

1. (f, g) is a strongly epimorphic pair;
2. $[f, g]$ is a strong epimorphism.

Proof. (1 $\implies$ 2) (f, g) is a strongly epimorphic pair. Consider the commutative diagram where m is a mono

$$\begin{array}{ccc} A + A' & \xrightarrow{[f,g]} & B \\ \downarrow l & & \downarrow h \\ C & \xrightarrow{m} & D \end{array}$$

Composing with the injection $\iota_1 : A \rightarrow A + A'$ and $\iota_2 : A' \rightarrow A + A'$, we have the following commutative diagram

$$\begin{array}{ccccc} & & B & & \\ & g \nearrow & & h \searrow & \\ A' & & A & & D \\ & \nwarrow \iota_2 & \searrow \iota_1 & & \\ & & C & & \end{array}$$

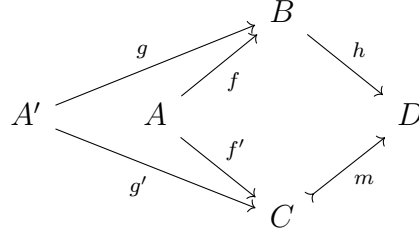
m (arrow from C to D)

Since (f, g) is strongly epimorphic, there exists a unique morphism $d : B \rightarrow C$ such that $\iota_1 d = f$, $\iota_2 d = g$ and $md = h$. The first two yield the equalities $d[f, g]\iota_1 = df = \iota_1 d$ and $[f, g]\iota_2 = dg = \iota_2 d$. So, by the universal property of the coproduct, we have that $d[f, g] = l$.

$$\begin{array}{ccccc} A & & & & A \\ \downarrow \iota_1 & & & & \downarrow \iota_1 \\ A + A' & \xrightarrow{[f,g]} & B & \xrightarrow{d} & C \\ \uparrow \iota_2 & & & & \uparrow \iota_2 \\ A' & & & & A' \end{array}$$

We already know that $md = h$, hence $[f, g]$ is a strong epimorphism.

(2 $\implies$ 1) $[f, g]$ is a strong epimorphism. Consider the following commutative diagram where m is a mono



Thus we have the commutative diagram

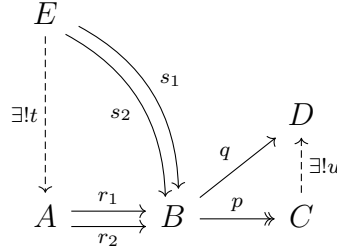
$$\begin{array}{ccc}
 A + A' & \xrightarrow{[f, g]} & B \\
 [f', g'] \downarrow & & \downarrow h \\
 C & \xrightarrow{m} & D
 \end{array}$$

Since $[f, g]$ is a strong epimorphism, there exists a unique morphism $d : B \rightarrow C$ such that $d[f, g] = [f', g']$ and $md = h$. Hence $f' = [f', g']\iota_1 = d[f, g]\iota_1 = df$ and $g' = [f', g']\iota_2 = d[f, g]\iota_2 = dg$. Thus (f, g) is a strongly epimorphic pair. $\square$

Proposition A.0.6. *If a regular epi has a kernel pair, then it is the coequalizer of its kernel pair.*

Proof. Let p be a regular epi, and let (r_1, r_2) be the kernel pair of p . Since p is a regular epi, it is the coequalizer of some parallel arrows. Let (s_1, s_2) be such pair as in the diagram below. We want to prove that p is the coequalizer of (r_1, r_2) .

We already know that $pr_1 = pr_2$, as (r_1, r_2) is the kernel pair of p . It remains to establish the universal property of the coequalizer. Let $q : B \rightarrow D$ be a morphism such that $qr_1 = qr_2$. By the universal property of the kernel pair, there exists a unique morphism $t : E \rightarrow A$ such that $r_1t = s_1$ and $r_2t = s_2$. We have that $qs_1 = qr_1t = qr_2t = qs_2$, so, by the universal property of the coequalizer for the pair (s_1, s_2) , there exists a unique morphism $u : C \rightarrow D$ such that $q = up$. Thus the universal property of the coequalizer also holds for the pair (r_1, r_2) .



□

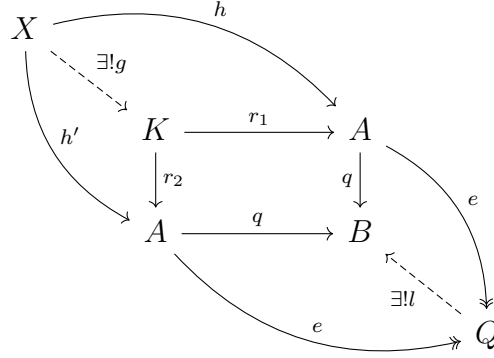
Proposition A.0.7. *If a kernel pair has a coequalizer, then it is the kernel pair of its coequalizer.*

Proof. Let (K, r_1, r_2) be a kernel pair of some arrow $q : A \rightarrow B$, and assume that such kernel pair has a coequalizer $e : B \rightarrow Q$. We want to prove that (r_1, r_2) is the kernel pair of its coequalizer e .

$$\begin{array}{ccccc}
 & & & & Q \\
 & & & \nearrow e & \\
 K & \xrightarrow[r_2]{r_1} & A & \xrightarrow{q} & B
 \end{array}$$

We already know that $er_1 = er_2$, since e is the coequalizer of (r_1, r_2) . It remains to show that the universal property of the kernel pair holds. Consider $h : X \rightarrow A$ and $h' : X \rightarrow A'$ morphisms such that $eh = eh'$. Since (r_1, r_2) is the kernel pair of q , it holds that $qr_1 = qr_2$. Thus by the universal property of the coequalizer e , there exists a unique morphism $l : Q \rightarrow B$ such that $q = le$. Composing with such morphism l we have that $leh = leh'$, hence $qh = qh'$. Since (r_1, r_2) is the kernel pair of q , by the universal property there exists a unique morphism $g : X \rightarrow K$ such that $r_1g = h$ and $r_2g = h'$. Hence (r_1, r_2) is the kernel pair of e , its

coequalizer.



□

Proposition A.0.8. *A regular epi is an epi.*

Proof. Let $p : A \rightarrow B$ be a regular epi. By the definition, it is the coequalizer of some parallel arrows (s_1, s_2) . Suppose that f and g are two morphisms such that $fp = gp$. We aim to prove that $f = g$.

Since p is the coequalizer of (s_1, s_2) , then $ps_1 = ps_2$. Moreover we have that $fps_1 = fps_2 = gps_2 = gps_1$.

$$S \begin{array}{c} \xrightarrow{s_1} \\ \xrightarrow{s_2} \end{array} A \xrightarrow{p} B \begin{array}{c} \xrightarrow{f} \\ \xrightarrow{g} \end{array} C$$

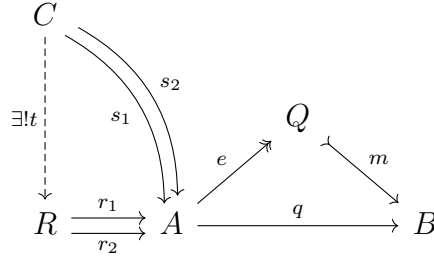
A dashed arrow $\exists!t$ points from B to C .

By the universal property of the coequalizer, there exists a unique morphism $t : B \rightarrow C$ such that $tp = fp = gp$. Hence, by uniqueness, $f = g$. □

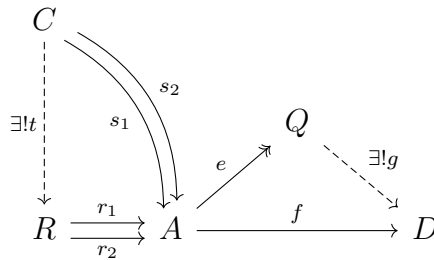
Proposition A.0.9. *In an exact category $\mathcal{A}$, each equivalence relation has a coequalizer and it is the kernel pair of its coequalizer.*

Proof. Consider an equivalence relation (R, r_1, r_2) in $\mathcal{A}$. Since $\mathcal{A}$ is exact, (R, r_1, r_2) is the kernel pair of some arrow $q : A \rightarrow B$. $\mathcal{A}$ is also regular, so q admits a factorization $q = me$, where m is a mono and e is a regular epi, ie the coequalizer of some arrows. Let s_1 and s_2 be such arrows, we have that $es_1 = es_2$. Composing with m , we have that $qs_1 = mes_1 = mes_2 = qs_2$. By the universal property of the kernel pair, there exists a unique morphism

$t : C \rightarrow R$ such that $r_1 t = s_1$ and $r_2 t = s_2$.



Now we want to prove that e is the coequalizer of (r_1, r_2) . We have that $mer_1 = fr_1 = fr_2 = mer_2$. Thus, since m is a mono, $er_1 = er_2$. It remains to prove the universal property of the coequalizer. Let $f : A \rightarrow D$ be a morphism such that $fr_1 = fr_2$. Composing with t we have $fr_1 t = fr_2 t$, hence $fs_1 = fs_2$. By the universal property of the coequalizer e , as coequalizer of s_1 and s_2 , there exists a unique morphism $g : Q \rightarrow D$ such that $ge = f$. Hence e is also the coequalizer of r_1 and r_2 .



Finally, we observe that (r_1, r_2) is the kernel pair of e , since by *Proposition A.0.7*, a kernel pair that admits a coequalizer is the kernel pair of its coequalizer. $\square$

Proposition A.0.10. *If a morphism f admits a factorization $f = mg$, with m mono, then f and g share the same kernel pair.*

Proof. Let (Eq, r_1, r_2) be the kernel pair of $f : A \rightarrow B$ and $g : A \rightarrow Q$, such that $f = mg$ with m mono. We have that $fr_1 = fr_2$, hence $mgr_1 = mgr_2$. Since m is a mono, we have that $gr_1 = gr_2$. Conversely, if (Eq, r_1, r_2) is the kernel pair of e , then we have that $gr_1 = gr_2$. Just composing with m

we have that $mgr_1 = mgr_2$. Hence $fr'_1 = fr'_2$.

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow g \quad \nearrow m & \\ & Q & \end{array}$$

Now we want to prove that if the universal property of the kernel pair holds for f , then it also holds for g . Let $s_1 : X \rightarrow A$ and $s_2 : X \rightarrow A$ be morphisms such that $gs_1 = gs_2$. Just composing with m , we have that $mgs_1 = mgs_2$. Hence $fs_1 = fs_2$. By the universal property of the kernel pair of f , there exists a unique morphism $k : X \rightarrow \text{Eq}$ such that $r_1k = s_1$ and $r_2k = s_2$. Conversely, consider $s_1 : X \rightarrow A$ and $s_2 : X \rightarrow A$ be morphisms such that $fs_1 = fs_2$, and suppose that the universal property holds for g . That is $mgs_1 = mgs_2$. Since m is a mono, we have that $gs_1 = gs_2$. By the universal property of the kernel pair of g , there exists a unique morphism $k : X \rightarrow \text{Eq}$ such that $r_1k = s_1$ and $r_2k = s_2$.

$$\begin{array}{ccccc} & & & & s_2 \\ & & & & \curvearrowright \\ X & & & & A \\ & \searrow k & & & \downarrow f \\ & \text{Eq} & \xrightarrow{r_2} & & B \\ & \downarrow r_1 & & & \downarrow m \\ & A & \xrightarrow{f} & & Q \\ & \searrow g & & & \uparrow \\ & & & & \end{array}$$

□

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